

WARNING TO PIRATES

Photocopying, reproduction, copying, or recording of the information contained in this book is highly prohibited. Going against these rules will result in expensive & dangerous consequences.



UNIVERSITY OF PORTHARCOURT



CAMPUS DREAM
POST UTME
PAST QUESTIONS &
ANSWERS

FOR

**SCIENCES, MEDICINE
& ENGINEERING**

2006 - DATE

TABLE OF CONTENTS

UNIPORT post-UTME 2025/2026	3 – 5
Solution to UNIPORT post-UTME 2025/2026	5
UNIPORT post-UTME 2024/2025	5 – 6
Solution to UNIPORT post-UTME 2024/2025	7
UNIPORT post-UTME 2023/2024	7 – 8
Solution to UNIPORT post-UTME 2023/2024	9
UNIPORT post-UTME 2018/2019	10 – 11
Solution to UNIPORT post-UTME 2018/2019	11
UNIPORT post-UTME 2017/2018	11 – 12
Solution to UNIPORT post-UTME 2017/2018	12
UNIPORT post-UTME 2015/2016	12 – 13
Solution to UNIPORT post-UTME 2015/2016	13
UNIPORT post-UTME 2014/2015	13 – 14
Solution to UNIPORT post-UTME 2014/2015	14
UNIPORT post-UTME 2013/2014	15 – 19
Solution to UNIPORT post-UTME 2013/2014	19
UNIPORT post-UTME 2012/2013	19 – 22
Solution to UNIPORT post-UTME 2012/2013	23
UNIPORT post-UTME 2010/2011	23 – 25
Solution to UNIPORT post-UTME 2010/2011	26
UNIPORT post-UTME 2009/2010	26 – 29
Solution to UNIPORT post-UTME 2009/2010	29
UNIPORT post-UTME 2008/2009	29 – 32
Solution to UNIPORT post-UTME 2008/2009	32
UNIPORT post-UTME 2007/2008	32 – 36
Solution to UNIPORT post-UTME 2007/2008	36

**HEY! DO YOU KNOW YOU CAN NOW WATCH THE LIVE TEACHING OF THE
SOLUTIONS OF UNIPORT POST-UTME QUESTIONS FROM 2006 – DATE?
OH YES ITS SIMPLE!**

**JUST USE YOUR INTERNET ENABLED MOBILE PHONE OR COMPUTER TO LOGIN TO
WWW.CAMPUSDREAM.COM.NG/SCHOOL/LOGIN.PHP**

**YOU WILL BE ASKED TO ENTER ‘ACCESS PIN’ ENTER THE
VISIT WWW.CAMPUSDREAM.COM.NG TO BUY THE ACCESS PIN OR CALL 08033924926**

First Published in 2014 by
Multi-purpose Publishers Nigeria
Suite 18, Abraham Plaza, Utako, Abuja
Revised 2015, 2016, 2017, 2018, 2019, 2023, 2024, 2025, 2026
Tel: 08033924926
Email: campusdreamnigeria@gmail.com

Website: www.campusdream.com.ng

**UNIVERSITY OF PORT HARCOURT
2025/2026 POST UTME SCREENING EXERCISE
CBT**

Choose the option that best conveys the meaning of the underlined words

1. Her summary of the meeting was brief and straight to the point. A. careful B. precise
C. accurate D. exact E. crucial
2. The whole students where happy that the academic year had taken o. A. gone o B.
started C. ended D. continued E. truncated
3. His penury gave him a life time of hunger. A. avariciousness B. a uence C. poverty D.
penny pinching E. greed

Choose the option that most suitably lls the space

4. He read more but he is tired A. ought to B. may C. would D. could E. should
5. We have been having series of meetings lately. A. some B. a C. a lot of D. many E. few
6. Don't forget to hang the telephone when you nish. A. down B. up C. in D. on E. out
7. Almost all building which in 1950s not habitable now. A. were build/are B. were
built/are C. were built/were D. were built/are E. are build/are
8. Some motor bikes Aroma and Unizik. A. shuttle along B. shuttle to C. shuttle between
D. shuttle from E. shuttle in
9. I you refuse the A. advice/advise B. advise/advice C. advise/advise D. advice/advice
E. adverse/adverse
10. The chairman, no less than the members to blame. A. is B. are C. were D. ought to
E. should
11. Did the ladies enjoy A. each other B. one another C. themselves D. theirselves
E. theirself
12. The worker did a shoddy job at that A. cite B. sight C. site D. sait E. side

Choose the option that is opposite in meaning to the underlined words

13. Their students are very conscientious. A. careful B. industrious C. carefree D. corrupt E.
energetic
14. The business is very unsuccessful. A. unscrupulous B. unskillful C. unlucky D. ungrateful
E. prosperous
15. Spurious arguments are easily identi ed by smart guys. A. genuine B. interesting C. false
D. illogical E. boring
16. In my opinion, the decision was disinterested. A. neutral B. impartial C. biased D. candid
E. foolish

Identify the word that has a di erent vowel sound from the others

17. A. key B. win C. even D. people
18. A. tear (verb) B. hear C. weird D. mere
19. A. cost B. won't C. hope D. coast

Identify the word that has a di erent consonant sound from the others

20. A. wife B. of C. o D. laugh

Choose the correct or best answer to the following questions

21. The lenticels: A. remove excess water in plant B. absorbs water from the atmosphere C. is for gaseous exchange D. absorbs light E. stores food
22. A photometer is used to determine the rate at which a shoot A. respire B. sucks air bubbles C. lose weight D. absorbs air E. transpires
23. One of these is not a digestive enzyme. A. bile B. lipase C. maltase D. pepsin E. ptyalin
24. Which of these is not metamerically segmented? A. tapeworm B. earthworm C. centipede D. crayfish E. shark
25. Which of these diseases could be exclusively associated with a river basin? A. malaria B. syphilis C. anchocerciasis D. cholera E. poliomyelitis
26. Which of these statements is true about enzymes? They A. are carbohydrates B. most times do not need co-factors to be active C. are not sensitive to hydrogen ion concentration D. are not specific in their actions E. cannot withstand high temperature
27. Soil erosion can be controlled by all these except A. planting cover crops B. contouring of sloping ground C. terracing poles D. laying of mulch E. all of the above
28. Evidence that the tooth is a living part of the body can be found in the A. gum B. pulp cavity C. cement D. enamel E. diastema
29. In higher animals, most cells retain power division except A. lymphocytes B. nerve cells C. malpighian cells D. germ cells E. liver cells
30. Taxism is different from tropism in that in taxism: A. the whole organism is affected B. one directional movement is involved C. it is a response to multi-directional movement D. part of the organism is affected E. the organism is not in motion
31. The molarity of a 20% by weight of aqueous solution of H_2SO_4 (molecular weight is 98) is
A. 4.55 B. 3.55 C. 2.55 D. 0.21 E. 0.51
32. A gas occupies 30.0dm^3 at s.t.p. what volume will it occupy at 91°C and 380mmHg ? A. 20.0dm^3 B. 40.0dm^3 C. 60.0dm^3 D. 80.0dm^3 E. 100.0dm^3
33. When ethane gas react with chlorine gas to form 1, 2-dichloroethane, the reaction is called? A. substitution B. addition C. polymerization D. hydrolysis E. reduction
34. The reaction of ethanoic acid and ethanol with acid catalyst will produce one of these as the organic product. A. ethyl ethanoate B. ethyl methyl ketone C. butanol D. chloroethane E. none of the above
35. In preparation of ammonia, the drying agent used is A. CaO B. CaCl_2 C. conc. H_2SO_4 D. P_2O_5 E. H_3PO_4
36. One of these has an isomer. A. C_2H_4 B. C_4H_{10} C. C_3H_8 D. C_2H_6 E. C_6H_8
37. In a flame test, copper exhibits what colour? A. green B. lilac C. blue-green D. crimson E. yellow-green
38. The characteristic reaction of the carbonyl compound is A. substitution B. elimination C. addition D. saponification E. oxidation – reduction
39. What volume of oxygen will remain after reacting 8cm^3 of hydrogen with 20cm^3 of oxygen?
A. 10.0cm^3 B. 12.0cm^3 C. 14.0cm^3 D. 16.0cm^3 E. 18.0cm^3
40. Which of these gases is the most dangerous pollutant? A. hydrogen sulphide B. carbon (IV) oxide C. sulphur (IV) oxide D. carbon (II) oxide E. all of the above

41. When the size of the hole of a pin-hole camera is increased, the result is A. a blurred image B. magnified image C. correction for chromatic aberration D. sharper focus of the image E. no much effect
42. The mechanical advantage of an inclined plane depends on: A. its length B. its height C. the product of its height and length D. the ratio of its height to its length E. the difference between its height and its length
43. On dispersing white light with spectrometer, the component with the shortest wavelength is A. orange B. green C. violet D. red E. yellow
44. One of these is a vector A. electric charge B. electric field C. electric potential difference D. electric capacitance E. electric resistance
45. Surface tension is important in A. the coating of a ship in water B. the coating of a dry needle in water C. the coating of a balloon in air D. the diffusion of a sugar solution across a membrane E. the diffusion of a digested food in animals
46. Thermionic emission is the process by which A. α -rays are produced B. x-rays are produced C. γ -rays are produced D. electrons leave hot element E. electrons are transferred to hot element
47. Conveyor belt have rough surface in order to A. make them carry load B. make them strong C. protect them when loads are on them D. make them slip less E. last longer
48. Most of the gas in a discharge tube is pumped out so that electricity is conducted at A. high pressure B. low pressure C. low voltage D. steady voltage E. high voltage
49. The ground is cold at night because A. sun no longer shines B. atmosphere reflects the sun's energy at night C. atmosphere absorbs the sun's energy at night D. earth radiates heat to the atmosphere at night E. there is dew at night
50. If the refractive index of a medium in air is 2.0. what is the critical angle for this medium?
A. 30° B. 42° C. 45° D. 50° E. 55°

SOLUTION TO UNIPORT 2025/2026 POSTUTME

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. B | 2. B | 3. C | 4. A | 5. B | 6. B | 7. B |
| 8. C | 9. B | 10. A | 11. C | 12. C | 13. C | 14. E |
| 15. A | 16. C | 17. B | 18. C | 19. A | 20. B | 21. C |
| 22. A | 23. A | 24. E | 25. C | 26. E | 27. D | 28. B |
| 29. D | 30. A | 31. A | | | | |

32. At s.t.p, $P_1 = 760\text{mmHg}$, $P_2 = 380\text{mmHg}$, $T_1 = 273\text{K}$, $T_2 = 91^\circ\text{C} = 91 + 273 = 364\text{K}$, $V_1 = 30.0\text{dm}^3$, $V_2 = ?$

From general gas law, $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$

$$\therefore V_2 = \frac{P_1 V_1 T_2}{T_1 P_2} = \frac{760 \times 30.0 \times 364}{273 \times 380} = 80.0\text{dm}^3 \quad \text{Ans: D}$$

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 33. A | 34. A | 35. A | 36. B | 37. E | 38. C | 39. D |
| 40. E | 41. A | 42. C | 43. C | 44. B | 45. A | 46. D |

47. D

48. B

49. D

50. Refractive index, $n = \frac{1}{\sin C} = \frac{1}{\sin 30^\circ}$

critical angle, c

$\sin C$

$\sin C = \frac{1}{2} = 0.5$; $C = \sin^{-1}(0.5) = 30^\circ$ Ans: A

technology D. green biotechnology

**UNIVERSITY OF PORT HARCOURT
2024/2025 POST UTME SCREENING EXERCISE
CBT**

INSTRUCTIONS: shade firmly only one answer using HB pencil only

Answer all questions

1. Our teacher was not speaking when he said that the lawyer was under a cloud.
A. literally B. simply C. frankly D. directly
2. The signature on the document lacked so, it was rejected.
A. clarity B. authenticity C. formality D. simplicity
3. Everyone could tell that Binta was happy from the on her face.
A. picture B. appearance C. expression D. sight
4. His last novel was by his visit to the prisons. A. inspired B. Generated
C. conceived D. revealed
5. As a result of the power cut, production at the paint factory came to a
A. hold-up B. blockade C. standstill D. suspense
6. Several were passed at the meeting. A. memoranda B. Decisions C. Resolutions
D. actions
7. James and Henry were at daggers drawn when I knew them. This means that they
A. were always angry with each other B. always wore daggers C. were good at drawing
daggers D. were fighting even / body
8. My book has become dog-eared through use. This means that A. the corners of the pages of
my book are crumbled B. my book is designed like dog's ears C. my book is torn D. my
book is indispensable
9. Musa does not believe that there is poverty in Africa; he is a dyed-in-the-wool capitalist. This
means that Musa is: A. a kind capitalist B. a thoughtful capitalist C. an old capitalist D. an
unbending capitalist.
10. Whenever my cousin comes to realize that I am telling the truth, she will eat humble pie. This
means that my cousin will A. respect me B. hate herself C. apologize D. be ashamed

From the words lettered A to D below in each of the following sentences, choose the word that is nearest in meaning to the underlined word as it is used in the sentence.

11. Cook had Oil and so could not prepare the soup.
A. little B. small C. some D. much
12. We will have to set up a Panel to investigate this matter.
A. seven men B. seven-men C. seven men D. seven-man
13. The workers could not move the heavy stone They tried.
A. how much B. as much as C. no matter that D. however hard
14. The boys football when the teacher entered the class.

- A. are discussing B. were discussing C. will be discussing D. will discuss
15. Our teacher is to make a mistake. A. too much intelligent B. Intelligent to much
C. much too intelligent D. to intelligent much

In each of the questions 6 – 10, select the option that best explains the information conveyed in the sentence

16. Okeke's chief idiosyncrasy is a passion for pounded yam. A. Okeke's chief hates pounded yam
B. Okeke hates pounded yam C. okeke has a penchant for pounded yam D. okeke's chief likes pounded yams
17. The conference is biennial. A. the conference is held twice every year B. the conference is held twice every two years C. the conference is held once every other year D. the conference lasts for two years.
18. Jimmy is as hard as nails. A. Jimmy is very strong B. Jimmy is very brave C. Jimmy is determined D. Jimmy is unsympathetic
19. The stadium had a capacity crowd in spite of the weather. A. the weather favoured the crowd at the stadium B. the crowd defied the weather C. the stadium wasnot filled because of the weather D. the crowd loved the weather at the stadium
20. Only two ministers survived the recent cabinet reshuffle. A. in the recent changes in government, only two ministers were sacked B. in the recent changes in government, all but two ministers remained alive C. in the recent changes in government, all but two ministers retained their posts D. in the recent changes in government, all but two ministers were sacked
21. A ball is thrown vertically upwards from the ground with an initial velocity of 50m/s. what is the total time spent by the ball in the air. A. 2.5s B. 5.0s C. 10.0s D. 15.0s E. 20.0s
22. A body of mass 60g appears to have a mass of 39g when totally immersed in oil and appears to have a mass 36g when totally immersed in water. Calculate the relative density of oil.
A. $\frac{8}{7}$ B. $\frac{13}{12}$ C. $\frac{12}{13}$ D. $\frac{7}{8}$ E. $\frac{13}{20}$
23. Two resistors R_1 and R_2 are connected in parallel, R_2 being greater than R_1 . The combined resistance is A. less than R_1 B. greater than R_2 C. the sum of R_1 and R_2 D. the difference of R_1 and R_2 E. greater than R_1 but less than R_2
24. Two capacitances of $6\mu\text{f}$ and $8\mu\text{f}$ are connected in series. What additional capacitance must be connected in series with this combination to give a total of $3\mu\text{f}$?
A. $3\mu\text{f}$ B. $16\mu\text{f}$ C. $24\mu\text{f}$ D. $30\mu\text{f}$ E. $14\mu\text{f}$
25. The upper and lower fixed points of thermometer T are 90 and 10 respectively. Determine the temperature on the Kelvin scale when the reading on T is 30^0 .
A. 25K B. 298K C. 80K D. 303K E. 293K
26. If a copper has a heat capacity of $80\text{J}^0\text{C}$ and specific heat capacity of 400J/kg . Calculate it's mass. A. 0.20kg B. 320kg C. 20.0kg D. 0.02kg E. 5kg
27. An object of height 12.0cm is placed from a pinhole camera, if the image distance from the pinhole is 10.0cm, calculate the image height.
A. 12.0cm B. 5.0cm C. 120.0cm D. 120.0m E. none of the above
28. Calculate the minimum value of the refractive index of a 45^0 prism which is used to turn a beam of light by total internal reflection through 90^0
A. 1 B. 1.50 C. 1.41 D. 1.20 E. 4.11
29. A wave of frequency 75Hz form a stationary pattern in a medium where the distance between adjacent nodes is 2.0cm. What is the velocity of the wave?
A. 3m/s B. 30m/s C. 300m/s D. 0.3m/s E. 150m/s
30. The pigment in the malphigian layer responsible for skin colouration is
A. haemoglobin B. haomocyanin C. chlorocruorin D. melanin
31. Cordination is achieved in the body of mammals through the action of the A. endocrine and nervous systems B. skeletal and nervous systems C. muscular and nervous systems
D. circulatory and nervous systems

32. Which of the following statements is correct of hormones? Hormones are
 A. cell into the blood through ducts B. secreted directly into the blood C. chemical substances in the blood stream D. non-specific in their mode of operation
33. The pituitary is called 'master gland' because A. it is located in the brain B. its lanes are more numerous than those of any other gland C. its secretion has other glands D. it is the only organ that produces hormone
34. Auxins are produced in the A. petiole of leaves B. parenchyma of roots and shoots C. epidermis of roots and shoot D. apical regions of root and shoots
35. The brain and the spinal cord make up the A. peripheral nervous system B. autonomic nervous system C. central nervous system D. somatic nervous system
36. Increasing specialization of structure and function among animals of the same class or order for survival in new and different environments is called?
 A. adaptive radiation B. phylogeny C. homology D. analogy
37. When large numbers of organisms share limited space and resources, result is
 A. immigration B. commensalism C. symbiosis D. competition
38. The change in colour of the chameleon serves as a means of A. attraction to the opposite sex B. repulsion of the enemy C. camouflage from a predator D. regulation of body temperature
39. Species can be defined as the group of organisms that A. resemble each and live in the same habitat B. are of common origin and are always found together C. resemble each other and can interbreed freely D. resemble each other and occupy the same niche
40. The process whereby a liquid turns spontaneously into vapour is called
 A. evaporation B. regelation C. boiling D. sublimation
41. The oxidizing agent commonly used for conversion of alcohol to aldehyde is A. potassium dichromate (IV) B. potassium dichromate (V) C. potassium dichromate (VI) D. potassium dichromate (VII) E. potassium dichromate (VIII)
42. Which one of the following is a hygroscopic compound? A. H_2SO_4 B. NaOH C. HNO_3 D. NaCl E. ECl
43. Which of these is constant R for ideal gas? A. 8.314Jkmol^{-1} B. $8.314\text{J}^{-1}\text{Kmol}^{-1}$ C. 8.314KJKmol D. $8.314\text{KJ}^{-1}\text{mol}^{-1}$ E. $8.314\text{JK}^{-1}\text{mol}^{-1}$
44. Which of these compounds is a covalent compound? A. ZnS B. NaCl C. CsCl D. SiO_2 E. TiO_2
45. Metallic elements usually have structure. A. open packed B. limited packet C. slight packet D. close packet E. ring packet
46. Find the value of x if $3(2^x) = 24$. A. 5 B. 2 C. 7 D. 8 E. 3
47. Find the value of K if the remainder when $3x^3 + Kx^2 - 11x + 12$ is divided by $x - 2$ is 16
 A. $-\frac{1}{2}$ B. $\frac{1}{2}$ C. $\frac{2}{3}$ D. $\frac{3}{2}$ E. $\frac{2}{5}$
48. If $a = \frac{1}{2-\sqrt{3}}$ and $b = \frac{1}{2+\sqrt{3}}$. Find the value of $a^2 + b^2$. A. 17 B. 14 C. 9 D. 7 E. 12
49. The product of 3 numbers in geometric progression is 720. Find the second term.
 A. 12 B. 7 C. 5 D. 9 E. 11
50. The sum of 3 numbers in arithmetic progression is 12 and the sum of their squares is 66. Find the possible value of the common difference.
 A. 2 or $\frac{1}{3}$ B. 3 or $\frac{3}{4}$ C. 5 or $\frac{3}{4}$ D. 3 or $\frac{5}{2}$ E. 2 or 7

SOLUTION TO UNIPOST 2024/2025 POSTUTME

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. D | 2. B | 3. C | 4. A | 5. C | 6. A | 7. A |
| 8. D | 9. D | 10. C | 11. A | 12. C | 13. D | 14. B |
| 15. A | 16. C | 17. D | 18. A | 19. B | 20. C | |
21. From $v = u + at$; $v = 0$, $u = 50\text{m/s}$, $g = 10\text{m/s}^2$
 $U = at$; $t = \frac{u}{g} = \frac{50}{10} = 5\text{s}$ Ans: B
22. Relative density of oil = $\frac{60 - 39}{60 - 36} = \frac{7}{8}$ Ans: D 23. A
24. Let the additional capacitance be x

For series connection: $\frac{1}{C} = \frac{1}{6} + \frac{1}{8} + \frac{1}{x} = \frac{1}{3}$

$$\frac{4x + 3x + 24}{24x} = \frac{1}{3}$$

$$3(7x + 24) = 24x; \quad 7x + 24 = 8x$$

$$8x - 7x = 24; \quad \therefore x = 24\mu\text{f} \quad \text{Ans: C}$$

25. $\begin{array}{c} 90 \\ | \\ 30 \\ | \\ 10 \end{array}$ $\begin{array}{c} 373 \\ | \\ X \\ | \\ 273 \end{array}$ $\frac{90 - 10}{30 - 10} = \frac{373 - 273}{X - 273}$
 $\frac{80}{20} = \frac{100}{X - 273}$
 $4(X - 273) = 100$
 $X - 273 = 25$
 $\therefore X = 25 + 273 = 298K$ Ans: B
26. Heat capacity, $H = mc$; $m = \frac{H}{c} = \frac{80}{400} = 0.20kg$ Ans: A
27. height of image = ?, height of object = 12cm, distance of image = 10cm, distance of object = 24cm
From Magnification = $\frac{\text{Height of image}}{\text{Height of object}} = \frac{\text{distance of image}}{\text{distance of object}} = \frac{x}{12} = \frac{10}{24}$
 $X = \frac{12 \times 10}{24} = 5cm$ Ans: B
28. Refractive index ${}_g\eta_a = \frac{\sin i}{\sin r} = \frac{\sin 45^\circ}{\sin 90^\circ} = 0.7071$
Also, ${}_a\eta_g = \frac{1}{{}_g\eta_a} = \frac{1}{0.7071} = 1.41$ Ans: C
29. distance between adjacent nodes = $\frac{\lambda}{2} = 2cm$
 $\lambda = 2 \times 2cm = 4cm = 0.04m$; $f = 75Hz$
but, velocity = $f\lambda = 75 \times 0.04 = 3m/s$ Ans: A
30. B 31. B 32. A 33. A 34. D 35. C 36. A
37. D 38. C 39. B 40. A 41. C 42. A 43. E
44. D 45. D
46. $3(2^x) = 24$; $2^x = \frac{24}{3} = 8$
 $2^x = 2^3$
 $\therefore x = 3$ Ans: E
47. $f(2) = 16 = 3(2)^3 + K(2)^2 - 11(2) + 12$
 $16 = 24 + 4k - 22 + 12$
 $16 = 4k + 14$
 $4k = 16 - 14 = 2$; $\therefore k = \frac{2}{4} = \frac{1}{2}$ Ans: B
48. $(\frac{1}{2-\sqrt{3}})(\frac{1}{2-\sqrt{3}}) + (\frac{1}{2+\sqrt{3}})(\frac{1}{2+\sqrt{3}})$
 $= \frac{1}{4-4\sqrt{3}+3} + \frac{1}{4-4\sqrt{3}+3} = \frac{1}{7-4\sqrt{3}} + \frac{1}{7+4\sqrt{3}} = \frac{14}{(7-4\sqrt{3})(7+4\sqrt{3})} = 14$ Ans: D
49. Let the numbers be a/r , a , ar ;
Product: $\frac{a}{r} \times a \times ar = 729$; $a^3 = 729$
 $\therefore a = 9$ Ans: D
50. ***

UNIVERSITY OF PORT HARCOURT
2023/2024 POST UTME SCREENING EXERCISE CBT

. USE OF ENGLISH - Comprehension

My good people: I come before you this evening as a man whose honesty and integrity have been questioned. Now, the usual political thing to do charges as leveled against you is either to ignore them or to deny them without giving details. But before I answer any of your question, let me state the categorically that i have not touched a kobo of the N5000,000 we contributed. Every kobo of it has been used in defraying political campaign expenses. As a matter of fact, during one of my meet the press conference, Mr. James ukpong accosted me and said, ' Honourable Senator what about this fund we hear about?' In answer to another one of hid questions, I said that neither contributors to this fund, nor contributors to any of my campaigns had ever received any special consideration that he would not have received as an ordinary

constituent. And I can say that never, since I become a senator, have I made a telephone call for them to an agency or have I made a telephone call for them to an agency or have I gone down to an agency on their behalf. Record will show that and these records are in the hands of the administrators

1. The author is
 - A. Exonerating himself from allegation of embezzlement
 - B. Ignoring the allegation of embezzlement
 - C. demonstrating that he is a senator
 - E. Trying to implicate ken kamalu, a fellow politician
2. Constituent in the passage means
 - A. A section of his constituency
 - B. His entire constituency
 - C. His campaigner
 - D. a person having voting rights where he is living
 - E. A supporter
3. Accosted in the passage means
 - A. Grossing insulted
 - B. Greeted
 - C. Fought with
 - D. Forced
 - E. Went and spoke to
4. It appears that the n500000
 - A. was paid into the government treasury
 - B. has been paid into the government treasury
 - C. will soon be paid into the government treasury
 - D. Has been saved by Ken Kamalu
 - E. Will be used for future campaigns
5. Ken Kamalu is portrayed as
 - A. a corrupt politician
 - B. the administrator of the fund
 - C. an ordinary taxpayer
 - D. a government official
 - E. a citizen

LEXIS AND STRUCTURE

In questions 6 to 9 choose the words opposite in meaning to the word underlined

6. He loved his altruism A. Benevolence B. Sincerity C. Selfishness D. Selflessness E. kindness
7. Disgruntled people are indifferent to any plans to rid the society of evil.
Different from B. Different about. C. In agreement with D. Interested in E. Opposed to
8. This card entitles you to attend the film show. A. Disqualifies B. Discourages. C. Disenchants
D. proclaims E. satisfies

9. Kelechi hardly ever falls sick A. Most often B. Very seldom C. Sometimes. D. Frequently E. Occasionally
10. Little Ugochi has a bicycle, which is in marvelous condition
A. Need of a tune up B. Excellent C. Questionable D unstable E. Working
11. Many state practice the ideals of democracy A. Government by elections
B. Government by representatives C. Government by civilians D. government by all people E. Oligarchy
12. The principal advised that we pursue this case cautiously, otherwise we are bound to be taken by Ada's lies A. Away B. Out C. in D. Off E. On
13. Samuel; would you please come here, john? John: no I'm busy In my farm.
A. I am working B. I'm working C. I am to work D. I shall have worked E. I go to work
14. James reminding that not all that glitter is gold.
A. Needs B. Need C. Needing D. Needs to E. Need to
15. The plane overshot the In a minor accident
A. Railway B. Hanger C. Tarmac –D. Runaway E. Road

BIOLOGY

1. In mammals, the function of the sebaceous gland is to A. produce sweat
B. secrete sodium C. secrete water D. produce an oil substance E. manufacture vitamin
2. Movement and positions of the head in man are detected by the
A. cochlea B. malleus C. utriculus D. semicircular canals E. outer
3. The maize grain is a fruit and not a seed because it has A. large endosperm
B. is formed from an ovary C. is a monocotyledon D. has no plumule and radical
E. has a hypogeal germination
4. The characters by which an organism is recognized are termed its
A. phenotype B. genotype C. morphology D. anatomy E. physiology
5. Germination which results in the cotyledons being brought above ground is called
A. hypocotyls B. epicotyls C. epigeal D. hypogeal E. plumule
6. In a mammal, stimulus is transferred from the receptor muscle to the central nervous system through the A. motor neurons B. effector muscle C. dentrities D. sensory neurons
E. synapses
7. Which of the following food chains is the correct sequence?
A. weeds tadpoles beetles fish man B. weeds tadpoles fish beetles man
C. tadpoles beetles weeds man fish D. man fish beetles tadpoles weeds
E. fish beetles tadpoles weeds tadpoles
8. The primary and secondary hosts respectively of bilharzias are A. fish and man B. man and dog
C. snail and man D. man and snail E. fish and snail
9. The origin of mineral particles in the seed is A. humus B. water C. micro-organisms
D. weathered rock E. organic matter

10. The initial volume of water poured in bag of dry soil was 50ml and the amount that drained through was 35ml. the percentage water content of the full soak soil is therefore A. 46.7
B. 25.0 C. 20.0 D. 30.0 E. 58.3
11. From the following list of type of mutation, identify the one that is not hereditary
A. germinal mutation B. somatic mutation C. germinal mutation D. genetic mutation
12. In a cell, digestive enzymes mostly occur in A. ribosome's B. lysosomes C. mitochondria
D. plastids
13. Which of the following habitats from the highest diversity of living species?
tropical rain forests B. savannah grassland C. desert D. tropical forests
14. Lack of protein in the diet of children manifest easily because
A. children do not store up protein efficiently
B. it is difficult for the children to chew meat
C. protein supplies energy to the body
D. protein is responsible for growth and repairs
15. Growth response of a plant to a light gradient is known as A. mastic movement B. geotropism C. hydrotropism D. Phototropism

PHYSICS

1. A body falls freely under gravity ($g = 9.8\text{m/s}^2$) from a height of 10m on top of a platform 0.8m above the ground. Its velocity on reaching the platform is
A. 7848m/s B. 80m/s C. 78m/s D. 27.78m/s
2. A hydrometer is an instrument used for measuring the A. the depth of water in a vessel
B. relative density of a liquid by method of flotation C. relative density of a liquid by finding the apparent loss in weight D. relative humidity of the atmosphere
3. A bead traveling on a straight wire is brought to rest at 0.2m by friction. If the mass of the bead is 0.01kg and the coefficient of friction between the bead and the wire is 0.1. Determine the work done by friction. A. $2 \times 10^{-4}\text{J}$ B. $2 \times 10^{-3}\text{J}$ C. $2 \times 10^1\text{J}$ D. $2 \times 10^2\text{J}$
4. A machine whose efficiency is 60% has a velocity ration of 5. if a force of 500N is applied to lift a load of P(N), what is the magnitude of P?
A. 750N B. 4166N C. 500N D. 1500N
5. A mass of gas at 7°C and 70cm of mercury has a volume of 1200cm^3 . determine its volume at 27°C and a pressure at 75cm of mercury A. 1200cm^3 B. 1378cm^3 C. 4320cm^3 D. 4629cm^3
6. A motor tyre is inflated to pressure of $2.0 \times 10^5\text{N/m}^2$ when the temperature of air is 27°C . What will be the pressure at 87°C assuming the volume does not change?
A. $2.6 \times 10^{-5}\text{N/m}^2$ B. $2.4 \times 10^{-5}\text{N/m}^2$ C. $2.2 \times 10^{-5}\text{N/m}^2$ D. $1.3 \times 10^{-5}\text{N/m}^2$
7. A beam of light is incident from air to water at an angle of 30° . Find the angle of refraction if the refractive index of water is $\frac{4}{3}$. A. 15° B. 18° C. 22° D. 240°
8. The wavelength of signal from a radio transmitter is 1500m and the frequency is 200 KHz. What is the velocity of the propagation? A. $3 \times 10^8\text{m/s}^2$ B. $7 \times 10^3\text{m/s}^2$ C. $3 \times 10^4\text{m/s}^2$ D. 7m/s^2

9. A boy on looking into a mirror discovers that his face appeared to have grown bigger. The boy must have been looking at a A. convex mirror with his face at the focus B. concave mirror with his face between the focus and the mirror C. convex mirror with his face between the focus and the mirror D. concave mirror with his face at the focus
10. Find the frequencies of the first three harmonics of a piano string of length 5m, if the velocity of the string is 120m/s. A. 10Hz, 80Hz, 120Hz B. 180Hz, 360Hz, 540Hz C. 80Hz, 160Hz, 240Hz D. 360Hz, 180Hz, 90Hz
11. The resistance of a piece of wire of length 20cm and cross-sectional area and resistivity is A. 1.0Ω B. 10.0Ω C. 400.0Ω D. $1.0 \times 10^{-13}\Omega$
12. An electric device is rated 2000W, 250V. The current fuse rating of the device is A. 8A B. 9A C. 7A D. 6A
13. Determine the inductive reactance when a 30.0mH inductor with negligible resistance is connected to a 1.3KHz oscillator. A. 39.0Ω B. 122.5Ω C. 245.0Ω D. 39KΩ
14. The half-life of a radioactive element is 9days. Calculate the fraction that remains after 36days? A. $\frac{1}{32}$ B. $\frac{1}{16}$ C. $\frac{1}{4}$ D. $\frac{15}{32}$
15. The graphite rods surrounding the uranium fuel rods in a nuclear reactor are used to absorb the neutrons and hence halt the nuclear process B. create the neutrons and hence slow down the nuclear process C. slow down the neutrons and hence slow down the nuclear process D. speed up the neutrons and hence speed up the nuclear process

CHEMISTRY

1. A certain volume of a gas at 298k is heated such that its volume and pressure are now four times the original values. What is the new temperature? A. 18.6k B. 100.0k C. 298.0k D. 1192.0k E. 4768.0k
2. The boiling points of water, ethanol, toluene and butan - 2 -ol are 373.ok, 352.3k, 383.6k and 372.5k respectively. Which liquid has the highest vapour pressure at 323.ok? A. water B. toluene C. ethanol D. butan-2-ol E. none
3. The function of sulphur during the vulcanization of rubber is to: A. act as catalyst for the polymerization of rubber molecules B. convert rubber from thermosetting to thermoplastic polymer C. from chains which bind rubber molecules together D. break down rubber polymer molecule E. shorten the chain length of rubber polymer
4. Complete hydrogenation of ethyne yields A. benzene B. methane C. ethane D. ethane E. propane
5. Mixing of aqueous solution of barium hydroxide and sodium tetraoxocarbonate (iv) yields a white precipitate of A. barium oxide B. sodium tetraoxocarbonate C. sodium oxide D. sodium hydroxide E. barium tetraoxocarbonate
6. An organic compound decolorized acidified-KMnO₄ solution but failed to react with ammoniacal silver nitrate solution. The organic compound is likely to be A. a carboxylic acid B. an alkane C. an alkene D. an alkyne E. an alkanone
7. Which of the following conducts electricity? A. sulphur B. graphite C. diamond

- D. red phosphorus E. yellow phosphorus
8. Which of the following compounds is not the correct product formed when the parent metal is heated in air? A. calcium oxide (CaO) B. sodium oxide (Na₂O)
C. copper (ii) oxide (CuO) D. tri-iron tetroxide (Fe₃O) E. aluminium oxide (Al₂O₃)
 9. When marble is heated to 1473 K, another whiter solid is obtained which reacts vigorously with water to give an alkaline solution. The solution contains
A. NaOH B. KOH C. Mg(OH)₂ E. Ca(OH)₂
 10. Which of the following roles does sodium chloride play in soap preparation? It
A. react with glycerol B. purifies the soap C. accelerates the decomposition of fat and oil D. separates the soap from the glycerol E. converts the fat acid to its sodium salt
 11. Which of the following is used as a moderator to control nuclear fission?
A. lead B. heavy water C. iron D. chromium
 12. One of the active components of baking powder is
A. MgSO₄ B. NaHCO₃ C. CaSO₄ D. NaCl
 13. H₂SO₄ is used to remove rust on the surface of iron (pickling) before electroplating. The type of reaction involved is A. redox reaction B. neutralization C. double decomposition D. hydrolysis
 14. Silver is chloride turned gray when exposed to sunlight because
A. the silver iron is reduced to silver B. the silver iron is oxidized to silver C. silver is a transition metal D. the silver chloride forms complexes in the sun
 15. Which of these compounds exhibits resonance?
A. benzene B. ethanol C. propene D. Butyne

MATHEMATICS

1. A rectangular lawn has an area of 1815 square yards. If its length is 50 meters, find its width in meters. Given that 1 meter equals 1.1 yards
A. 39.93 B. 35.00 C. 33.00 D. 30.00
2. A man invests a sum of money at 4% per annum simple interest. After 3 years, the Principal amount to N7,000.00. Find the sum invested.
A. N7,840.00 B. N6,250.00 C. N6,160.00 D. N5,8333.33
3. By selling 20 oranges for N1.35 a trader makes a profit 8%. What is his percentage gain or loss if he sells the same 20 oranges for N1.10?
A. 8% B. 10% C. 12% D. 15%
4. Four boys and ten girls can cut a field in 5 hours. If the boys work at $\frac{1}{4}$ the rate of which the girls work, how many boys will be needed to cut the field in 3 hours?
A. 180 B. 60 C. 25 D. 20
5. The minimum value of y in the equation $y = x^2 - 6x + 8$ is
A. 8 B. 3 C. 0 D. -1
6. A room is 12m long, 9m wide and 8m high. Find the cosine of the angle which a diagonal of the room makes with the floor of the room
A. $\frac{15}{17}$ B. $\frac{8}{17}$ C. $\frac{8}{15}$ D. $\frac{12}{17}$
7. The base of a pyramid is a square of side 8cm. If its vertex is directly above the center, find the height, given the edge is 4√3 cm
A. 6cm B. 5cm C. 4cm D. 3cm

8. Find the probability of selecting a figure which is parallelogram from a square, a rectangle, a rhombus, a kite and a trapezium A. $\frac{3}{5}$ B. $\frac{2}{5}$ C. $\frac{4}{5}$ D. $\frac{1}{5}$?
9. The numbers 3, 2, 8, 5, 7, 12, 9 and 14 are the marks scored by a group of students in a class test. If p is the mean and q the median then p + q is
A. 18 B. $\frac{17}{12}$ C. 16 D. 15
10. The goals scored by 40 football teams from three league divisions are recorded below
- | | | | | | | | |
|---------------|---|---|----|----|---|---|---|
| no. of goals: | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
| Frequency: | 4 | 3 | 15 | 16 | 1 | 0 | 1 |
- What is the total number of goals scored by all the teams?
A. 21 B. 40 C. 91 D. 96
11. If the third and sixth terms of a GP are 12 and 96 respectively, find the sum of the first four terms of the GP.
A. $\frac{45}{2}$ B. 45 C. 24 D. 48
12. If $\frac{dy}{dx} = \sin x + 2x - 1$ and $y(0) = 2$, find $y(x)$.
A. $\cos x + x^2 - x - 3$ B. $\cos x + x^2 - x - 1$ C. $\cos x + x^2 - x + 1$ D. $\cos x + x^2 - x + 3$
13. A sector of a circle of radius 30cm which subtends an angle 120° at the centre is used to form a cone. Find the radius of the cone
A. 10.3cm B. 15.3cm C. 10cm D. 15cm
14. The organizers of a national sports fiesta want to choose four sponsors from seven banks and five telecom companies. In how many ways can they choose the sponsors if they must include at least one bank and at least one telecom company?
A. 455 B. 495 C. 490 D. 40
15. Evaluate $\frac{(0.6744)^2 - (0.3256)^2}{0.05}$
Correct to three significant figures
A. 69.8 B. 6.97 C. 6.98 D. 6.976

AGRICULTURAL SCIENCE

1. A characteristic feature of subsistence agriculture is
A. supply of labour by farm families
B. wholesale of farm harvest
C. purchasing of seeds for planting
D. disease's control
E. none of the above
2. Which of the following is primary product of the forest trees?
A. paper
B. particle board
C. timber
D. cardboard
E. pulp
3. The components of a complete farm enterprise record include the following except
A. list of receipts and expenses of the enterprise
B. list of payable and receivable accounts
C. farm income statements
D. list of marketing centres
E. an inventory
4. One of the sexually transmitted diseases in livestock is
A. brucellosis
B. rinderpest
C. scabies
D. ring worm
E. red water

5. Checking on costs and returns enables the farmer to A. ascertain the immediate financial position of the farm B. ascertain the future financial position of the farm C. keep farm records D. plan the organization of a farm annually E. know when there is excessive depreciation of farm machinery
6. Exposing farmers to scientific knowledge is the responsibility of the A. research institute B. farm settlers C. agro-service centre D. extension personnel E. agricultural development project
7. Stud bull is the term used in describing A. a castrated male cattle B. a bull kept for breeding C. an old bull used for draft purposes D. a bull used for artificial insemination E. an immature male cattle
8. Which of the following groups of crops will perform better in the savannah zone of West Africa? A. cotton, millet and groundnut B. cotton, coffee and cocoa C. tobacco, sorghum and banana D. sorghum, millet and oil palm E. millet, melon and cassava
9. The following livestock diseases are caused by viruses except A. anthrax B. foot and mouth diseases C. rinderpest D. Newcastle disease E. fowl pox
10. Investments with long life span in agricultural economics are otherwise known as A. labour B. entrepreneurship C. saving deposit D. profit E. capital
11. The most important soil micro organisms concerned with decay and nutrient recycling are A. bacteria and fungi B. fungi and algae C. algae and actinomyces D. actinomyces and bacteria E. bacteria and algae
12. When leaves of crop plants show brownish colouration from the tips of the mid-ribs backward, they are showing deficiency symptom of A. magnesium B. zinc C. nitrogen D. copper E. phosphorus
13. Which of the following crops cannot be easily improved by cross-breeding A. groundnut B. cowpea C. oil palm D. maize E. yam
14. The correct sequence in preparing new land for cropping is usually A. stumping, laying out of farm, ridging and tilling B. laying out of farm, ridging, stumping and tilling C. tilling, stumping, ridging and laying out of farm D. ridging, laying out of farm, tilling and stumping E. laying out of farm, stumping, tilling and ridging
15. Diarrhea tainted with blood is a symptom of A. trypanosomiasis B. coccidiosis C. ketosis D. brucellosis E. Aspergilliosis

GEOGRAPHY

1. If the distance between two points on a map with scale 1:100,000 is 35mm, what is the distance between them on the ground? A. 3.00km B. 3.10km C. 3.59km D. 3.70km
2. Planetary winds are deflected on crossing the equator due mainly to the effects of A. gravitational force B. corollas force C. the jet stream D. frictional force
3. Two examples of inland drainage lakes in Africa are: A. lake Chad and lake Turkana B. lake Malawi and lake Turkana C. lake Chad and lake Edwards D. lake Victoria and Lake Ngami
4. The main use of the Abney Level is to determine the A. spot height B. relative location C. angle of inclination D. area coverage

5. A map measures 10cm by 6cm. if the map is enlarged such that its area is 240cm^2 , the map is enlarged
A. five times B. three times C. twice D. four times
6. Exfoliation is a weathering process commonly found in what region?
A. desert B. forest C. savannah D. tundra
7. Which of the following is the major factor responsible for Japan's unparalleled industrial growth in the last decade? A. availability of crude oil B. a large population
C. technological proficiency D. Japan's proximity to mainland Asia
8. The wind system associated with harmattan in West Africa is referred to as:
A. southwest trades B. Northeast trades C. Haze D. southeast trades
9. The location of most of the light manufacturing industries in African is determined primarily by the location of A. raw materials B. markets C. railway terminals D. labour
10. It is generally believed that there is a general rise in sea levels as a result of
A. global warming B. excessive rainfall C. excavation of sand from ocean beds
D. man-induced coastal erosion
11. If a map has a scale of 1:50,000 and a cassava farm is represented on the map by a rectangle 4cm by 5cm, what is the area of the farm?

A. 5km B. 6km C. 20km D. 25km
12. Which of the following sets of rocks is organically formed? A. clay, mudstone, sandstone
B. rock salt, breccias, conglomerate C. peat, chalk, limestone D. grit, coal, shale
13. When a river capture occurs, the behead stream is called the A. pirate B. gap C. elbow
D. misfit
14. Vegetation regeneration on a fallow land is a typical example of A. rejuvenation
B. reforestation C. vegetation conservation D. secondary succession
15. Which of the following groups of factors is most likely to influence industrial location?
A. Capital, raw material and large population B. raw material, transport and migration C. raw material, energy and market D. market, migration and labour

SOLUTION TO UNIPOST 2023/2024 POST UTME

USE OF ENGLISH

- | | | | | | | |
|------|------|-------|-------|-------|-------|-------|
| 1. A | 2. D | 3. E | 4. A | 5. B | 6. C | 7. D |
| 8. A | 9. B | 10. B | 11. B | 12. C | 13. B | 14. A |

15 D

BIOLOGY

- | | | | | | | |
|------|------|------|------|------|------|------|
| 1. D | 2. D | 3. B | 4. A | 5. C | 6. D | 7. B |
| 8. D | 9. E | | | | | |

10. Initial volume = 50m; Drained volume = 35m

Remaining volume = 50m - 35m = 15ml

Percentage water content = $\frac{\text{remaining vol}}{\text{Initial vol}} \times 100 = \frac{15}{50} \times 100$

$$= \frac{15}{50} \times 100 = 30\% \quad \text{Ans. D } 30\%$$

11. B 12. B 13. A 14. D 15. D

PHYSICS

1. $v^2 = u^2 + 2gs$ where $u = 0$, $g = 9.8\text{m/s}^2$, $s = 10\text{m}$

$$v^2 = 0 + 2(9.8 \times 10) = 196$$

$$\therefore v = \sqrt{196} = 14\text{m/s} \text{ [we neglect the height of the platform from the ground]}$$

2. B

3. $W = \mu mg \times s$ [where $\mu mg = F$]

$$= 0.1 \times 0.01 \times 9.8 \times 0.2 = 2 \times 10^{-3}\text{J} \quad \text{Ans: B}$$

4. $\text{efficiency} = \frac{L_E}{VR} \times 100\%$

$$60/100 = P/500/5$$

$$\therefore P = 0.6 \times 5 \times 500 = 1500\text{N} \quad \text{Ans: D}$$

5. $T_1 = 7^\circ\text{C} + 273 = 280\text{K}$, $T_2 = 27^\circ\text{C} + 273 = 300\text{K}$, $P_1 = 70\text{cmHg}$, $P_2 = 75\text{cmHg}$

$$\text{From } \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$V_2 = \frac{P_1 V_1 T_2}{P_2 T_1} = \frac{70 \times 1200 \times 300}{75 \times 280} = 1200\text{cm}^3 \quad \text{Ans: A}$$

6. $P_1 = 2.0 \times 10^5\text{N/m}^2$, $P_2 = ?$, $T_1 = 27^\circ\text{C} + 273 = 300\text{K}$, $T_2 = 87^\circ\text{C} + 273 = 369\text{K}$

$$P_1/T_1 = P_2/T_2$$

$$P_2 = \frac{P_1 \times T_2}{T_1} = \frac{2.0 \times 10^5 \times 360}{300} = 2.4 \times 10^5\text{N/m}^2, \quad \text{Ans: B}$$

7. $n = \sin i / \sin r = 4/3$

$$\sin r = \frac{3}{4} \times \sin 30 = \frac{3}{4} \times \frac{1}{2} = \frac{3}{8}$$

$$\therefore r = \sin^{-1}(3/8) = 22^\circ \quad \text{Ans: C}$$

8. From $v = f\lambda$ [where $v = ?$, $f = 200\text{KHz} = 200 \times 1000 = 200000\text{Hz}$, $\lambda = 1500\text{m}$]
 $1500 = 3 \times 10^8\text{m/s} \quad \text{Ans: A}$

$$V = 200000 \times$$

9. B 10. B

11. $R = \frac{\rho l}{A} = \frac{4 \times 10^{-7} \times 0.2}{8 \times 10^{-5}} = 1.0 \times 10^{-13} \Omega \quad \text{D}$

12. Power, $P = IV$;

$$I = \frac{P}{V} = \frac{2000}{250} = 8\text{A} \quad \text{Ans: A}$$

13. reactance of inductor, $X_L = 2\pi fL$ [$L = 30\text{mH} = 30 \times 10^{-3}\text{H}$, $F = 1.3\text{KHz} = 1.3 \times 1000\text{Hz}$ $X_L = 2 \times \frac{22}{7} \times 1.3 \times 1000 \times 30 \times 10^{-3} = 245.0\Omega$ Ans: A 14. B 15. C **See MultiPurpose Calculations In Physics By J. O. Onuoha For More Problems & Explanations**

CHEMISTRY

1. $\frac{V_1 P_1}{T_1} = \frac{V_2 P_2}{T_2}$

Were $V_1 = 1\text{dm}^3$ (given); $P_1 = 1\text{atm}$ (given); $T_1 = 298\text{k}$

$$V_2 = 4 \times V_1 = 4\text{dm}^3 \text{ (given)}$$

$$P_2 = 4 \times P_1 = 4\text{atm} \text{ (given)}$$

$$T_2 = ?$$

$$T_2 = \frac{V_2 P_2 \times T_1}{V_1 \times P_1} = \frac{4 \times 4 \times 298}{1 \times 1} = 4768\text{k} \text{ Ans: E}$$

2. ans. C ethanol
 3. ans. B from chains which bind rubber molecules together. 4. E
 5. ans. E $\text{Ba(OH)}_2 + \text{Na}_2\text{CO}_3 \rightarrow \text{BaCO}_3 + \text{NaOH}$
 6. Alkenes decolorize KMnO_4 but do not give metallic derivative test.
 7. ans. B graphite conducts electricity
 8. ans. D tri-iron tetra-oxide $[\text{Fe}_3\text{O}_4]$ is not formed differently by burning iron in the air.
 9. CaCO_3 (marble) $\rightarrow \text{CaO} + \text{CO}_2$
 $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$ Ans. E
 10. ans. D - separates the soap from the glycerol 11. ans. B - heavy water
 12. ans. B NaHCO_3 (baking soda) 13. ans. A redox reaction
 14. ans. C silver is a transition 15. ans. A benzene exhibits resonance

AGRICULTURAL SCIENCES

- | | | | | | | |
|-------|------|-------|-------|-------|-------|-------|
| 1. A | 2. C | 3. D | 4. | 5. A | 6. D | 7. B |
| 8. A | 9. A | 10. E | 11. E | 12. B | 13. E | 14. E |
| 15. A | | | | | | |

GEOGRAPHY

- | | | | | | | |
|-------|------|-------|-------|-------|-------|-------|
| 1. C | 2. B | 3. A | 4. C | 5. D | 6. C | 7. C |
| 8. B | 9. A | 10. A | 11. A | 12. C | 13. D | 14. B |
| 15. C | | | | | | |

2022/2023 POST UTME SCREENING EXERCISE CBT

Choose the option that best conveys the meaning of the underlined words

1. Her summary of the meeting was brief and straight to the point. A. careful B. precise C. accurate D. exact E. crucial
2. The whole students where happy that the academic year had taken o. A. gone o B. started C. ended D. continued E. truncated
3. His penury gave him a life time of hunger. A. avariciousness B. a uence C. poverty D. penny pinching E. greed

Choose the option that most suitably lls the space

4. He read more but he is tired A. ought to B. may C. would D. could E. should
5. We have been having series of meetings lately. A. some B. a C. a lot of D. many E. few
6. Don't forget to hang the telephone when you nish. A. down B. up C. in D. on E. out
7. Almost all building which in 1950s not habitable now. A. were build/are B. were built/are C. were built/were D. were built/are E. are build/are
8. Some motor bikes Aroma and Unizik. A. shuttle along B. shuttle to C. shuttle between D. shuttle from E. shuttle in
9. I you refuse the A. advice/advise B. advise/advice C. advise/advise D. advice/advice E. adverse/adverse
10. The chairman, no less than the members to blame. A. is B. are C. were D. ought to E. should
11. Did the ladies enjoy A. each other B. one another C. themselves D. theirselves E. theirself
12. The worker did a shoddy job at that A. cite B. sight C. site D. sait E. side

Choose the option that is opposite in meaning to the underlined words

13. Their students are very conscientious. A. careful B. industrious C. carefree D. corrupt E. energetic
14. The business is very unsuccessful. A. unscrupulous B. unskillful C. unlucky D. ungrateful E. prosperous
15. Spurious arguments are easily identi ed by smart guys. A. genuine B. interesting C. false D. illogical E. boring
16. In my opinion, the decision was disinterested. A. neutral B. impartial C. biased D. candid E. foolish

Identify the word that has a di erent vowel sound from the others

17. A. key B. win C. even D. people
18. A. tear (verb) B. hear C. weird D. mere
19. A. cost B. won't C. hope D. coast

Identify the word that has a di erent consonant sound from the others 20.

- A. wife B. of C. o D. laugh

Choose the correct or best answer to the following questions

21. The lenticels: A. remove excess water in plant B. absorbs water from the atmosphere C. is for gaseous exchange D. absorbs light E. stores food
22. A photometer is used to determine the rate at which a shoot A. respire B. sucks air bubbles C. lose weight D. absorbs air E. transpires
23. One of these is not a digestive enzyme. A. bile B. lipase C. maltase D. pepsin E. ptyalin
24. Which of these is not metamerically segmented? A. tapeworm B. earthworm C. centipede D. cray sh E. shark
25. Which of these diseases could be exclusively associated with a river basin? A. malaria B. syphilis C. anchocerciasis D. cholera E. poliomyelitis
26. Which of these statements is true about enzymes? They A. are carbohydrates B. most times do not need co-factors to be active C. are not sensitive to hydrogen ion concentration D. are not specific in their actions E. cannot withstand high temperature
27. Soil erosion can be controlled by all these except A. planting cover crops B. contouring of sloping ground C. terracing poles D. laying of much E. all of the above
28. Evidence that the tooth is a living part of the body can be found in the A. gum B. pulp cavity C. cement D. enamel E. diastema
29. In higher animals, most cells retain power division except A. lymphocytes B. nerve cells C. malpighian cells D. germ cells E. liver cells
30. Taxism is different from tropism in that in taxism: A. the whole organism is affected B. one directional movement is involved C. it is a response to multi-directional movement D. part of the organism is affected E. the organism is not in motion
31. The molarity of a 20% by weight of aqueous solution of H_2SO_4 (molecular weight is 98) is A. 4.55 B. 3.55 C. 2.55 D. 0.21 E. 0.51
32. A gas occupies 30.0dm^3 at s.t.p. what volume will it occupy at 91°C and 380mmHg? A. 20.0dm^3 B. 40.0dm^3 C. 60.0dm^3 D. 80.0dm^3 E. 100.0dm^3
33. When ethane gas react with chlorine gas to form 1, 2-dichloroethane, the reaction is called? A. substitution B. addition C. polymerization D. hydrolysis E. reduction
34. The reaction of ethanoic acid and ethanol with acid catalyst will produce one of these as the organic product. A. ethyl ethanoate B. ethyl methyl ketone C. butanol D. cholera E. none of the above
35. In preparation of ammonia, the drying agent used is A. CaO B. CaCl_2 C. conc. H_2SO_4 D. P_2O_5 E. H_3PO_4
36. One of these has an isomer. A. C_2H_4 B. C_4H_{10} C. C_3H_8 D. C_2H_6 E. C_6H_8
37. In a ame test, copper exhibits what colour? A. green B. lilac C. blue-green D. crimson E. yellow-green
38. The characteristic reaction of the carbonyl compound is A. substitution B. elimination C. addition D. saponification E. oxidation – reduction
39. What volume of oxygen will remain after reacting 8cm^3 of hydrogen with 20cm^3 of oxygen? A. 10.0cm^3 B. 12.0cm^3 C. 14.0cm^3 D. 16.0cm^3 E. 18.0cm^3

40. Which of these gases is the most dangerous pollutant? A. hydrogen sulphide B. carbon (IV) oxide
C. sulphide (IV) oxide D. carbon (II) oxide E. all of the above
41. When the size of the hole of a pin-hole camera is increased, the result is A. a blurred image
B. magnified image C. correction for chromatic aberration D. sharper focus of the image
E. no much effect
42. The mechanical advantage of an inclined plane depends on: A. its length B. its height C. the product of its height and length D. the ratio of its height to its length E. the difference between its height and its length
43. On dispersing white light with spectrometer, the component with the shortest wavelength is A. orange B. green C. violet D. red E. yellow
44. One of these is a vector A. electric charge B. electric field C. electric potential difference D. electric capacitance E. electric resistance
45. Surface tension is important in A. the coating of a ship in water B. the coating of a dry needle in water C. the coating of a balloon in air D. the diffusion of a sugar solution across a membrane E. the diffusion of a digested food in animals
46. Thermionic emission is the process by which A. α -rays are produced B. x-rays are produced
C. γ -rays are produced D. electrons leave hot element E. electrons are transferred to hot element
47. Conveyor belt have rough surface in order to A. make them carry load B. make them strong C. protect them when loads are on them D. make them slip less E. last longer
48. Most of the gas in a discharge tube is pumped out so that electricity is conducted at A. high pressure B. low pressure C. low voltage D. steady voltage E. high voltage
49. The ground is cold at night because A. sun no longer shines B. atmosphere reflects the sun's energy at night C. atmosphere absorbs the sun's energy at night D. earth radiates heat to the atmosphere at night E. there is dew at night
50. If the refractive index of a medium in air is 2.0. what is the critical angle for this medium?
A. 30° B. 42° C. 45° D. 50° E. 55°

SOLUTION TO UNIPORT 2022/2023 POSTUTME

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. B | 2. B | 3. C | 4. A | 5. B | 6. B | 7. B |
| 8. C | 9. B | 10. A | 11. C | 12. C | 13. C | 14. E |
| 15. A | 16. C | 17. B | 18. C | 19. A | 20. B | 21. C |
| 22. A | 23. A | 24. E | 25. C | 26. E | 27. D | |

28. B

29. D

30. A

31. A

32. At s.t.p, $P_1 = 760\text{mmHg}$, $P_2 = 380\text{mmHg}$, $T_1 = 273\text{K}$, $T_2 = 91^\circ\text{C} = 91 + 273 = 364\text{K}$, $V_1 = 30.0\text{dm}^3$, $V_2 = ?$

From general gas law, $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$

$$\therefore V_2 = \frac{P_1 V_1 T_2}{P_2 T_1} = \frac{760 \times 30.0 \times 364}{380 \times 273} = 80.0\text{dm}^3 \quad \text{Ans: D}$$

$T_1 P_2 = 273 \times 380$

33. A

34. A

35. A

36. B

37. E

38. C

39. D

40. E

41. A

42. C

43. C

44. B

45. A

46. D

47. D

48. B

49. D

50. Refractive index, $n = \frac{1}{\sin C} = \frac{1}{\sin 30^\circ}$

critical angle, $C = \sin^{-1}(0.5)$

$\sin C = \frac{1}{n} = 0.5$; $C = \sin^{-1}(0.5) = 30^\circ$ Ans: A

technology D. green biotechnology

UNIVERSITY OF PORT HARCOURT 2018/2019 POST UTME SCREENING EXERCISE CBT

Choose the one that is nearly opposite in meaning to the underlined word and which will at the same time correctly fill the gap

- The handsome profit from her fish business made Mrs. Uba stop complaining about his husband salary. a. ugly b. meager c. modest d. lowly
- It pays to be firm rather than in a case like this. a. flexible b. frank c. irresponsible d. careless
- Usually Joseph is articulate in his speech but today he sounds a. clear b. turgid c. timid d. incoherent
- The man's first wife is loyal, while the second is a. unfaithful b. unkind c. selfish d. Hostile
- Instead of reconciling her parents, Roxanne worked hard at them a. dividing b. destroying c. separating d. Isolating
- We thought that uncle Sam's health had improved but it rather a. persisted b. deteriorated c. deepened d. decreased

From the words lettered A to D, choose the word that best completes each of the following sentences

- One very important aspect of human relationship is trust a. communal b. individual c. mutual d. personal
- The two tasks were performed and we were able to finish early a. alternatively b. simultaneously c. consecutively d. Separately

9. The committee is charged with finding a solution to the crisis
a. lengthy b. current c. lasting d. Strong
10. We are worried by the That exists between the two factions of the party
a. distraction b. commotion c. confusion d. rivalry
11. Which path does sound entering the human ear follow?
A. oral window ossicles eardrum
B. eardrum oval window ossicles
C. ossicles eardrum oval window
12. The activity of ptyalin is likely to decrease with an increase in the concentration of
A. oxygen B. starch C. protein D. acid
13. An individual in the AB blood group is a universal recipient for blood done because
A. all the red blood cells do not carry antigen B. all the red blood cells carry antigen A
C. all the red blood cells carry antigen B D. there are no antigen A and B anti-bodies in the serum
14. Neurons that receive stimuli from the body or internal organs are called A. sensory neurons
B. efferent neurons C. motor neurons D. relay neurons
15. A population is defined as a collection of similar organism that A. behave in the same way
B. interbreed freely C. are found in the same habitat D. eat the same food
16. In a typical predator food chain involving secondary and tertiary consumers, organism become progressively A. smaller and more numerous along the food chain
B. equal in number and size along the food chain C. larger and fewer along the food chain
D. parasitized along the food chain as consumers get bigger
17. In a bean seed, absorption of water at the beginning of germination is through the a. Hilum
b. Microphyle c. Testa d. plumule
18. The reproductive system of a male mammal is made up of a. Claspers, prostate gland, sperm duct and vas deferens b. Testes prostate gland, sperm duct and vas deferens c. Oviduct, urethra, testis and sperm duct d. Testis, uterus, prostate gland and sperm duct
19. The axial skeleton is found in the a. Skull, ribs, vertebral column and breast bone b. Skull, humerus, vertebral column c. Breast bone, Clavicle, ribs and vertebral d. Femur, sternum, ulna and skull
20. In insects, the structure that performs the same function as the kidney in man is the a. Nephridium b. Flame cell c. Malpighian tubule d. Trachea
21. 120cm^3 of Hydrogen were sparked with 60cm^3 of Oxygen at 110°C . what was the volume of stream produced? The equation for the reaction is $2\text{H}_{2(g)} + \text{O}_{2(g)} \rightarrow 2\text{H}_2\text{O}_{2(g)}$.
A. 30cm^3 B. 120cm^3 C. 60cm^3 D. 180cm^3
22. The number of orbital in a p-sub level of an atom is A. 2 B. 3 C. 5 D. 6
23. The number of sulphur atoms in 3.20g of sulphur IV oxide $\text{SO}_{2(g)}$ is (O = 16.0; S = 32.0; Avogadro constant = 6.02×10^{23}) A. 3.01×10^{22} B. 6.02×10^{22} C. 6.02×10^{23} D. 1.20×10^{24}
24. The shape of a graphite crystal is A. Tetrahedral B. Pyramidal C. Octahedral D. Hexagonal
25. The following substances are ores of metals except
A. bauxite B. cuprite C. graphite D. cassiterite
26. Which of the following processes does not involve the use of limestone A. Extraction of iron in the blast furnace B. Production of cement C. Production of washing soda by Solvay process D. Manufacture of Tetraoxosulphate (VI) acid by contact process.
27. Which of the following separation techniques would show that black ink is a mixture of chemical compounds? A. chromatography B. crystallization C. filtration D. sublimation
28. The compound that makes palm wine taste sour after exposure to the air for few days is
A. Ethanol B. Ethanoicacid C. methanol D. methanoicacid
29. 35cm^3 of hydrogen was sparked with 12cm^3 of oxygen at 110°C and 760mmHg to produce steam. What percentage of the total volume of gas left after the reaction is hydrogen?

- A. 11% B. 31% C. 35% D. 69% E. 80%
30. In the extraction of iron in the blast furnace, lime stone is used to
 - A. release CO_2 for the reaction
 - B. reduce the iron ore
 - C. increase the strength of the iron
 - D. remove impurities
 31. A sealed flask contains 600cm^3 of air at 27°C . It is heated to 35°C at a constant pressure. The new volume of air is.
 - A. 508cm^3
 - B. 516cm^3
 - C. 608cm^3
 - D. 603cm^3
 - E. 616cm^3
 32. A real image three times the size of an object is formed 24cm from a converging mirror. What is the focal length of the mirror?
 - A. 6cm
 - B. 8cm
 - C. 12cm
 - D. 16cm
 - E. 24cm
 33. An electrical device is rated 1500W and its resistance is 375Ω . The current it will draw is
 - A. 0.10A
 - B. 4.00A
 - C. 0.40A
 - D. 77.50A
 - E. 2.00A
 34. Three inductors of inductances 5mH , 10mH and 20mH are connected in series. The effective inductance is:
 - A. 2.90mH
 - B. 0.35mH
 - C. 3.5mH
 - D. 35.00mH
 - E. 29.0mH
 35. The amount of energy released when 0.5kg of uranium is burnt completely is
 - A. $4.5 \times 10^{16}\text{J}$
 - B. $1.5 \times 10^{16}\text{J}$
 - C. $1.5 \times 10^8\text{J}$
 - D. $4.5 \times 10^{18}\text{J}$
 - E. $4.5 \times 10^8\text{J}$
 36. A convex lens of focal length 10.0cm is used to form a real image which is half the size of the object. How far from the object is the image?
 - A. 45cm
 - B. 30cm
 - C. 15cm
 - D. 20cm
 37. A diverging lens of focal length 20cm forms an image half the size of the object. What is the object distance?
 - A. 11.11cm
 - B. 100cm
 - C. 60cm
 - D. 8.71cm
 38. An object of height 3.00cm is placed 10cm in front of a biconvex lens of focal length 15cm . The image of the object is
 - A. real and 3.00cm tall
 - B. virtual and 3.00cm tall
 - C. virtual and 9.00cm tall
 - D. real and 9.00cm tall
 39. The most suitable type of mirror used for the construction of a searchlight is the
 - A. concave mirror
 - B. spherical mirror
 - C. convex mirror
 - D. parabolic mirror
 40. Light waves and ripples of water are similar because both
 - A. are longitudinal waves
 - B. can be diffracted and refracted
 - C. have the same velocity
 - D. have the same frequency
 41. A binary operation $*$ on the set of rational numbers is defined as $x * y = 2x + (x^2 - y^3)/x + y$; find $-1 * 2$
 - A. 11
 - B. -11
 - C. 8
 - D. -8
 42. A polynomial in x whose zeroes are 2 , 1 and -3 is
 - A. $x^3 - 7x + 6 = 0$
 - B. $x^3 + 7x - 6 = 0$
 - C. $x^3 - 7x - 6 = 0$
 - D. $x^3 + 7x + 6 = 0$
 43. Find the range of values of x for which $7x - 3 > 3x + 4$
 - A. $x < \frac{7}{4}$
 - B. $x > \frac{7}{4}$
 - C. $7x < 4$
 - D. $-4x < 7$
 44. Let P be a probability function on set S , where $S = \{C_1, C_2, C_3, C_4\}$. Find $P(C_3)$ if $P(C_1) = \frac{3}{10}$ and $P(C_4) = \frac{1}{4}$
 - A. $\frac{2}{5}$
 - B. $\frac{1}{2}$
 - C. $\frac{1}{6}$
 - D. $\frac{1}{3}$
 45. Calculate the standard deviation of the following data $7, 8, 9, 10, 11, 12, 13$.
 - A. 2
 - B. 4
 - C. 10
 - D. 11
 46. A bricklayer measured the length of a wall and obtained 4.10m . If the actual length of the wall is 4.25m , find his percentage error.
 - A. $\frac{39}{17}\%$
 - B. $\frac{337}{41}\%$
 - C. 15%
 - D. $\frac{358}{17}\%$
 47. Factorize the expression: $2y^2 + xy - 3x^2$
 - A. $2y(y + x) - 3x^2$
 - B. $(2y - x)(2y + x)$
 - C. $(3x - 2y)(x - y)$
 - D. $(2y + 3x)(y - x)$
 48. What must be added to the expression $x^2 - 18 - x$ to make it a perfect square?
 - A. 3
 - B. 9
 - C. 36
 - D. 81
 49. The angle of a sector of a circle of radius 8cm is 240° . This sector is bent to form a cone. Find the radius of the base of the cone.
 - A. $\frac{18}{3}\text{cm}$
 - B. $\frac{15}{3}\text{cm}$
 - C. $\frac{16}{3}\text{cm}$
 - D. $\frac{8}{3}\text{cm}$
 50. The curved surface area of a cylindrical tin is 704cm^2 , calculate the height when the radius is 8cm . (Take $\pi = \frac{22}{7}$)
 - A. 3.5cm
 - B. 7cm
 - C. 14cm
 - D. 28cm

SOLUTION TO UNIPORT 2018/2019 POSTUTME CBT

1. B 2. A 3. D 4. C 5. C 6. B 7. C 8. B

9. C 10. D 11. B 12. D 13. D 14. B 15. C
 16. C 17. C 18. B 19. A 20. C 21. B 22. B 23. D
 24. D 25. C 26. D 27. A 28. B

29. Reaction equation: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$

Reacting volume: 35cm^3 12cm^3

Expected volume: 24cm^3 12cm^3

Excess gas = $35 - 24 = 11\text{cm}^3$

% of total volume of gas left = $\frac{11}{35} \times 100\% = 31\%$ Ans: B

30. D

31. $V_1 = 600\text{cm}^3$, $T_1 = 27^\circ\text{C} = 300\text{K}$, $T_2 = 35^\circ\text{C} = 308\text{K}$, $V_2 = ?$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\therefore V_2 = \frac{V_1 \times T_2}{T_1} = 616\text{cm}^3$$

32. B

33. $P = 1500\text{W}$, $R = 375\Omega$

But $P = I^2 R$

$$I^2 = \frac{1500}{375} = 4$$

$$I = \sqrt{4} = 2\text{A} \quad \text{Ans: E}$$

34. A 35. A

36. C 37. C 38. C 39. D 40. B

41. $x * y = 2x + \frac{x^3 - y^3}{x + y}$ 42. A

$$-1 * 2 = 2(-1) + \frac{(-1)^3 - (2)^3}{-1 + 2} = -2 + \frac{(-1 - 8)}{1}$$

$$= -2 - 9 = -11 \quad \text{Ans: B}$$

43. $7x - 3 > 3x + 4$ [collecting like terms]

$$7x - 3x > 3 + 4$$

$$4x > 7 \quad \therefore x > \frac{7}{4} \quad \text{Ans: B}$$

44. $P(C_3) = 1 - [P(C_1) + P(C_2) + P(C_4)] = 1 - [\frac{3}{10} + \frac{1}{3} + \frac{1}{5}]$ 45. A

$$= 1 - \frac{5}{6} = \frac{1}{6} \quad \text{Ans: C}$$

46. % error = $\frac{\text{error}}{\text{Actual length}} \times 100\% = \frac{4.25 - 4.10}{4.25} \times 100\%$

$$= \frac{0.05}{4.25} \times 100\% = \frac{39}{17} \quad \text{Ans: A}$$

47. $2y^2 + xy - 3x^2 = (2y^2 + 3xy)(2xy - 3x^2)$

$$= y(2y + 3x) - x(2y + 3x) = (yx)(2y + 3x) \quad \text{Ans: D} \quad 48. B$$

49. since the length of arc is bent to form a cone

$$\therefore 2\pi R = \frac{\theta}{360} \times 2\pi r$$

Where R = radius of base of the cone

$$\therefore R = \frac{\theta}{360} \times r = \frac{240}{360} \times 8 = \frac{16}{3} \text{ cm} \quad \text{Ans: A}$$

50. Curved surface area of a cylinder = $2\pi rh = 704$

$$H = \frac{704}{16\pi} = 14\text{cm} \quad \text{Ans: C}$$

SEE **MULTIPURPOSE**
CALCULATIONS IN
PHYSICS BY ENGR. J. O
 ONUOHA
FOR MORE PROBLEMS

UNIVERSITY OF PORT HARCOURT
2017/2018 POST UTME SCREENING EXERCISE CBT

Choose the preposition which best completes each sentence

1. Since childhood I have always had a passion music. A. on B. in C. for D. About
2. Parents punish their children because of the love they have ... them. A. on B. in C. for D. about
3. Her gown is as green as: A. grace B. grass C. flag D. butter

Complete the following simile:

4. Olu is as hungry as a: A. orphan B. wolf C. termites D. Thug
5. He is as mischievous as: A. Satan B. monkey C. gazelle D. tortoise
6. He is as industrious as: A. an ant B. elephant C. worker D. messenger

Choose a word opposite in meaning to the first word

7. Above: A. Over B. Below C. High D. Low
8. Liberty: A. Freedom B. Captivity C. Stadium D. Guilty
9. Dissent: A. Dissent B. Undissent C. Assent D. Indissent
10. Sharp: A. Clever B. Blunt C. Keen D. Even
11. Conductor: A. Driver B. Conductress C. Passenger D. Wire
12. Spendthrift: A. Miser B. Loan C. Foolish D. Happy
13. Profane: A. Beautiful B. Sacred C. Profuse D. Stiff.
14. She was the first woman to stand election to parliament. A. to B. for C. through D. by
15. If 1liter of 22M sulphuric acid is poured into a bucket containing 10liters of water, and the resulting solution 25cm^3 of 0.1M Na_2CO_3 to neutralize it. What is the original volume of the acid?
A. 2.2M B. 1.1M C. 0.22M D. 0.1 1M E. 0.20M
16. A solution of HCl contains 7.3g/dm^3 of the acid. What volume of it will neutralize 0.25M solution of Na_2CO_3 of volume 20ml A. 250ml B. 50ml C. 750ml D. 10ml E. 25ml
17. Which one of the following reactions is correct as written?
A. $\text{FeCl}_2 + \text{NH}_4\text{OH} \rightarrow \text{Fe}(\text{OH})_4 + \text{NHCl}$ B. $\text{Pb}_2\text{O}_3 + 4\text{HCl} \rightarrow \text{PbCl}_2 + 2\text{H}_2\text{O}$
C. $\text{PbNO}_3 + \text{H}_2\text{S} \rightarrow \text{PbS} + 2\text{HNO}_3$ D. $\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{H}_2\text{O}$
E. $2\text{NH}_4\text{NO}_3 \rightarrow \text{N}_2 + 2\text{H}_2\text{O}$
18. How many grams of hydrogen gas will be liberated when 6g of magnesium ribbon dissolves in 500cm^3 of 6M HCl? (Mg = 24, H =1, Cl = 35.5). A. 2.2g B. 0.7g C. 0.5g D. 0.3g E. 12g
19. An example of a strong electrolyte is A. ethyl acetate B. ethanol C. glucose D. sodium. Formate E. formic acid
20. The oxidation state of manganese in KMnO_4 is A. 44 B. 13 C. 15 D. 17 E. 12
21. The more the hydrogen peroxide (H_2O_2) in urine, the A. lighter the color of the strip B. the acidic the urine be C. darker the color of the strip D more healthier a person be
22. Hydrogen peroxide (H_2O_2) A. helps line up cell proteins B. is poisonous to tissues C. is often needed as a catalyst D. aids in oxidation of glucose
23. The nature of rennin is slightly A. Acidic B. alkaline C. water based D. mineral based
24. The application of biotechnology procedures in agricultural field is classified as
A. white biotechnology B. red biotechnology C. blue biotechnology D. green biotechnology
25. Consequences of water pollution includes A. discharge of untreated sewage B. burning of fossil fuels C. loss of habitat D. climate changes
26. Deforestation is caused due to A. Grazing B. timber C. Cultivation D. all of these
27. Endemics include A. Malaria B. plague C. cholera D. typhoid
28. In the outer corner of the upper lid, is the A. lachrymal tear gland B. naso-lachrymal duct C. liberal palpebral ligament D. orbital part
29. The innermost layer of the eyeball is A iris B. retina C. ciliary body D. pupil
30. Front of the ribs are attached to the A. diaphragm B. sternum C. external intercostal muscles D. internal intercostal muscles
31. Momentum is represented dimensionally as: (a). LT^{-2} (b). MLT^{-2} (c). MLT^{-1} (d). ML^{-1}T^2
32. Far point of vision is at: (a) infinity (b) 30cm (c) 25cm (d) 20cm

33. A car travelling at a constant speed of 20m/s, overcomes a constant frictional resistance of 300N, what is the H.P of the engine {take 1 HP = $\frac{3}{4}$ kw} (a) 7H.P (b) 10H.P (c) 5H.P (d) 8H.P
34. Using a converging lens, an object is placed at distance less than F from the lens, images are (a) virtual, upright and magnified (b) real, upright and diminished (c) virtual, real and magnified (d) at infinity
35. Suppose the angle of incidence in glass is C, so that the angle of refraction in air is 90° , if its glass has a refractive index of $\frac{3}{2}$ for light passing from air to glass. Calculate the critical angle (a) 40° (b) 52° (c) 38° (d) 42°
36. Which of the following waves is not transverse? (a) light waves (b) sound waves (c) sea waves (d) radio waves
37. Which of the following instruments is used to measure relative humidity? (a) Hydrometer (b) Barometer (c) Hypsometer (d) Hygrometer
38. The maximum displacement of a vibrating tuning fork is its (a) Amplitude (b) frequency (c) period (d) wavelength
39. An ideal step-down transformer steps up; (a) Power (b) energy (c) current (d) voltage
40. As a positively charged rod is brought nearer to the top cap of a positively charged gold leaf electroscope, the divergence of the leaves will A. decrease to zero B. steadily increase C. decrease to zero and then increase D. remain constant E. increase to a maximum and then decrease
- A bag contains 10 balls of which 3 are red and 7 are white. Two balls are drawn at random. Find the probability of none of the balls is red if the draw is
41. with replacement: A. 0.9 B. 1 C. 0.4 D. 0.49
42. Without replacement. A. 0.1 B. 0.47 C. 0.42 D. 0.21
43. A regular polygon has each of its angles as 160° . What is the number of sides of the polygon? A. 36 B. 9 C. 18 D. 20
44. A girl walks 30m from a point P on a bearing of 040° to a point Q. she then walks 30m on a bearing of 140° to a point R. the bearing of R from P is A. 90° B. 50° C. 45° D. 40°
45. How many different three digit numbers can be formed using the integers 1 to 6 if no integer occurs twice in a number? A. 24 B. 120 C. 60 D. 48
46. Find the 9th term of the arithmetic progression, 18, 12, 6, 0, 6 A. -54 B. -30 C. 30 D. 42 E. 54
47. Solve the equation: $\frac{4}{a} + \frac{1}{5a} = 3$ A. $2\frac{1}{2}$ B. $\frac{12}{5}$ C. $\frac{11}{3}$ D. $\frac{14}{15}$ E. $\frac{24}{5}$
48. What is the number whose logarithm to base 10 is 3.47712? A. 3.0 B. 0.3 C. 0.03 E. 0.00003
49. The nth term of a sequence is given by $(-1)^{n-2} 2^{n-1}$, find the sum of the second and third terms. A. -2 B. 1 C. 2 D. 6 E. -1
50. Evaluate $\log_{10}25 + \log_{10}32 - \log_{10}8$. A. 0.2 B. 2 C. 100 D. 409 E. 4

SOLUTION TO UNIPORT 2017/2018 POST-UTME

1. C 2. C 3. B 4. B 5. B 6. A 7. B 8. B
9. C 10. B 11. C 12. A 13. B 14. B 15. C

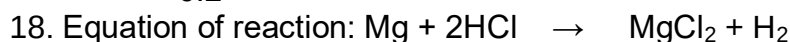


From neutralization equation: $C_A V_A = n_a$
 $C_B V_B = n_b$

$$C_A = 7.3\text{g/dm}^3 = 0.2\text{mol/dm}^3, V_A = ? C_B = 0.25\text{mol/dm}^3, V_B = 20\text{ml}, n_a = 2, n_b = 1$$

$$\frac{0.2 \times V_A}{0.25 \times 20} = \frac{2}{1}$$

$$V_A = \frac{0.25 \times 20 \times 2}{0.2} = 50\text{ml} \quad \text{Ans: B} \quad 17. D$$



SEE **MULTIPURPOSE**

CALCULATIONS IN

PHYSICS BY ENGR. J. O

ONUOHA

24g of Mg liberated 2g of H_2
 $\therefore$ 6g of Mg will liberate Xg of H_2
 $X = \frac{6 \times 2}{24} = 0.5g$ of H_2

24 **Ans: C** 19. E

20. The oxidation state of $KMnO_4$ is obtained thus

$$KMnO_4 = 1 + Mn + (-2 \times 4) = 0$$

$$1 + Mn = 8$$

$$Mn = 8 - 1 = +7$$

This is the electron in the outermost shell of manganese. Looking at the options you will observe that option D (i.e 17) has 7 electrons in its outermost shell.

Thus our answer is option D

- | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|
| 21. D | 22. A | 23. A | 24. C | 25. A | 26. B | 27. B | 28. A |
| 29. A | 30. C | 31. C | 32. A | 33. D | 34. A | 35. D | 36. B |
| 37. D | 38. A | 39. D | 40. B | | | | |

41. Since none of the two ball pick is red. Then they are both white

$$Pr(W) = \frac{7}{10} \times \frac{7}{10} = \frac{49}{100} = 0.49 \quad \text{Ans: D}$$

42. If the pick is done without replacement,

$$Pr(W) = \frac{7}{10} \times \frac{6}{9} = \frac{14}{30} = 0.47 \quad \text{Ans: B}$$

43. Each interior angle of a polygon is equal to $\frac{(n-2)180}{n}$

$$160 = \frac{(n-2)180}{n}$$

$$160n = 180n - 360; \quad 180n - 160n = 360$$

$$20n = 360$$

$$\therefore n = \frac{360}{20} = 18 \quad \text{Ans: C}$$

44. $q^2 = 30^2 + 30^2 - 2 \times 30 \times 30 \cos 80$
 $= 1800 - 312.5667 = 1487.4332$

$$q = \sqrt{1487.4332} = 38.5673m$$

$$\frac{30}{\sin \theta} = \frac{38.5673}{\sin 80}$$

$$\sin \theta = \frac{30 \sin 80}{38.5673} = 0.7660$$

$$\sin \theta = \frac{30 \sin 80}{38.5673} = 0.7660$$

$$\theta = \sin^{-1}(0.7660) = 49.99^\circ = 50^\circ$$

$$\text{bearing of R from P} = 40 + 50 = 90^\circ \quad \text{Ans: A}$$

45. The digit given are 1, 2, 3, 4, 5, 6. Since we have 6 digits to take three at a time,
The number of three digit form is ${}^6P_3 = \frac{6!}{(6-3)!} = \frac{6 \times 5 \times 4 \times 3!}{3!} = 120 \quad \text{Ans: B}$

$$(6-3)! \quad 3!$$

46. $a = 18, d = 6 - 12 = -6, n = 9$

$$T_n = a + (n-1)d$$

$$T_9 = 18 + (9-1)(-6) = 18 - 48 = -30 \quad \text{Ans: B}$$

47. $\frac{4}{a} + \frac{1}{5a} = 3$

$$\frac{(20+1)}{5a} = 3$$

$$21 = 15a$$

$$\therefore a = \frac{21}{15} = \frac{12}{5} \quad \text{Ans: B} \quad 48. E$$

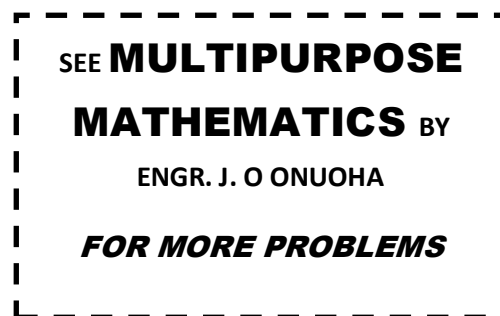
49. $n^{\text{th}} = (-1)^{n-2} 2^{n-1}$

$$T_2 = (-1)^{2-2} 2^{2-1} = 2$$

$$T_3 = (-1)^{3-2} 2^{3-1} = -1 \times 4 = -4$$

$$\text{Sum: } T_2 + T_3 = 2 + (-4) = 2 - 4 = -2 \quad \text{Ans: A}$$

50. $\log_{10} 25 + \log_{10} 32 - \log_{10} 8 = \log_{10} ({}^{25 \times 32}_{/8})$
 $= \log_{10} 100 = 2 \log_{10} 10 = 2(1) = 2 \quad \text{Ans: B}$



UNIVERSITY OF PORTHARCOURT
2014/2015 POSTUTME SCREENING EXERCISE
FACULTY OF HEALTH SCIENCES

1. A diverging lens forms an image of an object placed 10cm in front of it, at a point 6.5cm from the lens. What is the focal length of the lens? A. 18.6cm B. 16.5cm C. +3.9cm D. 21.2cm
2. Which colour of white light deviates most from the original path when light is incident on a piece of glass? A. red B. blue C. yellow D. orange
3. Which of the following statement correctly defines temperature? A. temperature causes hotness B. the total energy of a body is measured in temperature C. temperature is the property of the body which determines whether it is in thermal equilibrium with another body D. thermometers measure temperature
4. The following set of instrument is used for temperature measurement? A. alcohol thermometer, calorimeter and thermocouple B. pyrometer, thermocouple and clinical thermometer C. barometer, resistance thermometer and viscometer D. none of the above
5. A pin is placed at the bottom of a beaker 80mm high and completely filled with petrol, when viewed from the top, the pin appears to be at a depth of 60mm, what is the refractive index of the petrol? A. 1.80 B. 1.50 C. 1.33 D. 1.5
6. A converging lens can be used as a magnifying glass if the object is placed A. at F B. between F and the pole C. at the pole D. at 2F
7. A converging lens of focal length F, forms a real image whose size is the same as that of the object. The object distance is equal to A. 4F B. 2F C. F D. $\frac{F}{2}$
8. When the generation is required from an ac generator, the slip-rings are replaced by a commutator which is made up of A. a split ring B. two split-rings C. two copper brushes D. two insulating coils
9. In semiconductors the Hall Effect can be used to identify A. transistor B. the carrier either electrons or holes C. charge densities and concentrations D. the generation of emf
10. In transformers the primary and secondary coil are wound on a A. laminated hard iron core B. permanent magnet C. laminated soft iron core D. insulated magnet
11. The components of the circulatory system are heart, blood vessels A. veins B. arteries C. lymph D. plasma
12. The veins of the leaf are formed by the A. vascular bundles B. cambium cell C. palisade tissue D. spongy mesophyll
13. The unit of measurements are usually in A. millimeter and micrometer B. centimeter and nanometer C. micrometer and nanometer D. meter and decimeter
14. The two type of biogenetic tissues are A. phloem and parenchyma B. meristematic and non-meristematic C. meristematic and fruit D. flower and fruit
15. The surface of an alveolus in a mammals is well supplied with tiny blood vessels known as A. capillaries B. arteries C. arterioles D. venules
16. The stunted growth of a leguminous plant suffering from nitrogen deficiency may be corrected by inoculating the soil with A. saprophytic bacteria B. rhizobium C. denitrifying bacteria D. nitrosomonas
17. The gall bladder of a mammal has duct connected to the A. duodenum B. liver C. pancreas D. small
18. The aspect of ferns are dispersed by A. wind B. water C. insects D. explosive mechanism
19. The spores of ferns are dispersed by A. wind B. water C. insects D. explosive mechanism
20. The structures found only in plant cells are: A. cell membrane and cytoplasm B. chromatin and nucleolus C. cell and chloroplast
21. The federal government sent a troop of to the war-torn community.

- A. actors B. mobile police C. detective D. soldiers
22. A swarm of attacked the picnic makers. A. termites B. birds C. gorillas D. bees
23. The night watch man has a of arrows. A. bag B. cache C. quiver D. bundle
24. The huge drag-net enclosed a of fish. A. shoal B. school C. den D. flock
25. A child who hits smaller and weaker children is A. a rascal B. an imp C. a bully D. a tyrant
26. A chemical substances put on cuts to prevent infection is A. antidote B. antiseptic
C. disinfection D. toxin
27. A trick played on a victim to make other laugh is A. a joke B. a practical joke C. aril-fool
D. jesting
28. A person who is tall and tin is called A. bony B. lanky C. skinny D. stocky
29. Samson had great strength of body could hardly be heard. Give one word for this
A. maturity B. stamina C. untiredness D. strong
30. The lecturer's words were so low they could hardly be heard. Give one word for this.
A. incomprehensible B. unintelligible C. unheard of D. inaudible
31. Evaluate $21.05347 \div 1.6324 \times 0.43$ to 3 decimal places.
A. 20.351 B. 20.352 C. 20.980 D. 20.981
32. Evaluate $\frac{(0.142 \times 0.275)}{7(0.02)}$ correct to 3 decimal places
A. 0.033 B. 0.039 C. 0.308 D. 0.358
33. Simplify $7^{1/12} - 4^{3/4} + 2^{1/2}$ A. 4 B. $4^{1/6}$ C. $4^{5/6}$ D. $5^{1/6}$
34. A cylindrical tank has a capacity of 3080m^3 . What is the depth of the tank if the diameter of its base is 14m? A. 20m B. 22m C. 23m D. 25m
35. A sector of a circle of radius 7.2cm which subtends an angle 300° at the centre is used to form a cone. What is the radius of the base of the cone? A. 6cm B. 7cm C. 8cm D. 9cm
36. The chord ST of a circle is equal to the radius, r of the circle. Find the length of arc ST
A. $\frac{\pi}{2}$ B. $\frac{\pi}{3}$ C. $\frac{\pi}{3}$ D. $\frac{\pi}{12}$
37. A point P moves such that it is equidistant from the points Q and R. find QR when PR = 8cm and $\angle PRQ = 30^\circ$. A. 4cm B. $4\sqrt{3}\text{cm}$ C. 8cm^2 D. $8\sqrt{3}\text{cm}$
38. Evaluate $\frac{\frac{3}{8} \div \frac{1}{2} - \frac{1}{3}}{\frac{1}{8} \times \frac{2}{3} + \frac{1}{3}}$
A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{1}{2}$ D. 1
39. In the following balanced ionic equation, the value of A and B are:
 $\text{AMnO}_4 + \text{H}_2\text{SO}_3 \rightarrow \text{Mn}^{2+} + \text{SO}_4^{2-} + \text{BH}^+ + \text{H}_2\text{O}$
A. 3 and 3 B. 2 and 6 C. 2 and 4 D. 3 and 4
40. milliliters of water is needed to dilute 64.8ml of 0.15M HF to obtain 0.03M HF
A. 1000 B. 648 C. 324 D. 282
41. If 1.65g CaSO_4 salt is required to make a saturated solution with 12.4cm^3 of water at 25°C . calculate the solubility of the salt at 25°C . (molar mass of CaSO_4 in 130g/mol)
A. 0.98mol/dm^3 B. 1.96mol/dm^3 C. 0.4mol/dm^3 D. 0.28mol/dm^3
42. has no effect on the position of a system at equilibrium and helps equilibrium to be attained in a shorter time. A. free energy B. enthalpy C. entropy D. catalyst
43. Gregory concluded that iron (II) chloride is a substance,; since it forms solution on exposure to atmosphere. A. deliquescent B. efflorescent C. solubility D. hygroscopic
44. The decreasing order of the relative strength of these acids p. HNO_3 q. HBr r. HClO_4 and o. H_2S is A. $p>q>r>s$ B. $r>s>q>p$ C. $q>s>r>p$ D. $q>r>p>s$
45. The production of fruit juice at a free energy change of -345kJ/mol and equilibrium constant of 3.66×10^{12} occurred at a temperature of (gas constant = 8.314J/k/mol).
A. 273k B. 288k C. 200k D. 373k
46. Oxidation reaction is the process of the following except A. electron loss B. oxygen gain
C. hydrogen loss D. electron gain
47. If the free energy change ΔG of an equilibrium reaction is zero, the reaction will

- A. stop B. move to the right C. is at equilibrium D. move to the left
48. The ability of carbon to form bonds with other carbon atoms thereby giving many interesting compounds is known as: A. Polymerization B. Catenation C. Allotropy D. Isotopy
49. when an s-orbital of a carbon atom mixes with three p-orbital's, the hybrid orbital formed is
A. $2sp^3$ B. sp C. sp^3 D. sp^2 E. s^2p^3
50. In the sodium fusion test, nitrogen compounds are converted to: A. cyanides B. nitrides
C. ammonia D. sulphides

SOLUTION TO UNIPORT 2014/2015 POSTUTME EXAM

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. A | 2. | 3. C | 4. D | 5. C | 6. | 7. |
| 8. A | 9. B | 10. C | 11. D | 12. A | 13. C | 14. D |
| 15. A | 16. B | 17. B | 18. C | 19. A | 20. C | 21. D |
| 22. D | 23. C | 24. B | 25. C | 26. C | 27. | |

UNIVERSITY OF PORTHARCOURT
2013/2014 POSTUTME SCREENING EXERCISE
FACULTY OF HEALTH SCIENCES

Answer all questions

From the options lettered A-D, choose the question tag that best completes the sentence below

1. He goes to the farm, A. can he? B. can't he? C. does he? D. doesn't he?
2. The score of students in five courses are given as 70, 75, x, 80, 90. If the mean of the students' score is 78, find the value of x. A. 75 B. 85 C. 95 D. 120
3. The marks obtained by students in a Mathematics test are given below: 1, 3, 2, 2, 3, 4, 1, 5, 10, 11, 9, 8, 12, 14. If A is the mean and B is the median, then calculate A + B.
A. 11.67 B. 12.85 C. 10.57 D. 15.56

choose from the options lettered A-D, the word that most appropriately completes the sentence below

4. Good people are always ready to help those who are not as fortunate as
A. theirs B. them C. they D. us

Choose the correct option

5. In a box there are 10 T-shirts, 15 long sleeve, shirts and 25 gowns. If one attire is picked at random, what is the probability that the attire is either a T-shirt or a gown?
A. $\frac{7}{11}$ B. $\frac{11}{23}$ C. $\frac{7}{100}$ D. $\frac{7}{10}$
6. Diki drives to work in 40 minutes; she takes the same route to return home. If her average speed on her way home is half as her average speed on the trip to work, how much time does she spend driving on the round trip? A. 1 hour B. 1 hour, 20 minutes C. 1 hour, 40 minutes D. 2 hours

Fill in the blank spaces with the correct word or group of words from the option lettered A-D

7. That is not your book, it is A. Johns B. John's C. Johns' own D. for John's

From the options lettered A-E, choose the one that is nearest in meaning to the word underlined

8. Our school prefect is too officious and we all hate him because of his behavior
A. efficient B. efficacious C. overzealous D. active
9. A heating instrument has 12 resistors and they are placed vertically and they have uniform reducing power. The powers of first, second, third and fourth resistors are 300Ω, 260Ω, 220Ω and 180Ω respectively. Find the power of the resistor. A. 950Ω B. 960Ω C. 970Ω D. 960Ω
10. Find the probability of obtaining a multiple of 3 from a number chosen randomly by a student from the integers between 1 and 20 inclusive. A. $\frac{2}{11}$ B. $\frac{3}{7}$ C. $\frac{3}{10}$ D. $\frac{2}{9}$
11. The statement, "My father gave me this watch, re-written in the passive form becomes, I this watch by my father. A. had been given B. am given C. were given D. was given
12. When two or more independent clauses are joined together by using a co-ordinating conjunctions, the type of sentence formed is referred to as A. simple sentence B. compound sentence C. compound-complex sentence D. simple-compound sentence
13. In the sentence "he listed himself in the directory", the underlined word is an example of
A. relative pronoun B. personal pronoun C. reciprocal pronoun D. reflexive pronoun
14. The first and last term of an A.P. are 10 and 90, if the sum of terms is 750, the number of terms is used to find the common difference if the sum is 1400. A. 11.0 B. 10.90 C. 12.90 D. 5.7
15. A total of m matches are needed to fill 40 match boxes with the same number of matches in each box. How many matches are in each box? A. $\frac{40}{m}$ B. 40m C. $\frac{m}{4}$ D. $\frac{m}{40-m}$

From the options lettered A-D, choose the word that best completes the sentence

16. The robbers were subjected a thorough beating by the irate mob.
A. to B. by C. with D. for
17. He has stopped drinking A. does he? B. has he? C. hadn't he? D. hasn't he?
18. Every community from time immemorial must have had known and respected for their ability to guide and younger members of their community towards the of the desired goals of their group. A. teachers/direct/attainment B. priests/follow/enforcement
C. fathers/flog/creating D. coaches/frustrate/achievement
19. The word written in capital letters in the sentence "you have to put it IN their food", is an example of A. a coordinating conjunction B. a subordinating conjunction C. a metaphor
D. a verb
20. A girl is 6 years younger than her brother. The product of their age is 13. Find their ages
A. 15 years, 6 years B. 9 years, 3 years C. 12 years, 5 years D. 15 years, 9 years
21. If $a:b = 6:5$, find the value of $[\frac{4a+6b}{3a-b}] \div [\frac{4a}{5b}]$. A. -73 B. 73 C. -7.03 D. 7.03
22. A carton of hard drinks containing 20 bottles of gulder, 15 bottles of star and 5 bottles of hero. What is the probability of not choosing any of the bottles from the information given?
A. $\frac{122}{128}$ B. $\frac{125}{128}$ C. $\frac{128}{145}$ D. $\frac{145}{154}$
23. A student travelled for x hours at 5km/hr and for y hours at 10km/hr. the journey was 35km altogether, find x and y if the average speed for the journey was 7km/hr.
A. 3 hrs; 3 hrs B. 3 hrs; 2 hrs C. 1 hrs; 4 hrs D. 2 hrs; 3 hrs
24. In a poll, it was noticed that 22 people read biology text book and 37 people read physics text book. If 45 people read either physics or biology text book, find the probability of the people who read both text book. I.e. physics and biology. A. $\frac{14}{45}$ B. $\frac{15}{45}$ C. $\frac{16}{45}$ D. $\frac{17}{45}$
25. Which of the following is the youngest plant tissue? A. meristem B. phloem C. epidermis
D. xylem
26. Viruses differ from all forms of life because they A. have a thick cell wall B. feed on waste products of other organisms C. cause infectious diseases D. require other living cells to multiply
27. The endoplasm cells found in hydra are used for A. reproduction B. offence and defense
C. locomotion and nutrition D. food collection
28. Annelids differ from nematodes in that they A. exhibit bilateral symmetry B. are triploblastic C. are metamerically segmented D. possess complete digestive system
29. Which of the following perform similar functions? A. ascospores and ascocarp
B. antherozoids and rhizoids C. sorus and indusium D. strobili and inflorescence
30. In ferns, the sporophyte A. develops from a haploid zygote B. produce spores
C. is haploid and dependent on the gametophyte D. is haploid and independent of the gametophyte
31. The group of insects that undergoes metamorphosis is A. house flies, beetles and cockroaches B. cockroaches, grasshopper and bees C. house flies, beetle and butterflies
D. aphids, grasshoppers and butterflies
32. The nitrogenous substance that is excreted by birds in order to conserve water is
A. ammonia B. urea C. uric acid D. nitric acid
33. In mammals, the exchange of nutrients and metabolic products occurs in the
A. lymph B. lungs C. heart D. liver
34. The part of the stomach nearer the gullet is called the A. epiglottis B. cardiac sphincter
C. duodenum D. pyloric sphincter
35. A certain volume of a gas at 298k is heated such that its volume and pressure are now four times the original values. What is the new temperature?
A. 18.6k B. 100.0k C. 298.0k D. 1192.0k E. 4768.0k

36. The boiling points of water, ethanol, toluene and butan-2-ol are 373.K, 352.3K, 383.6K and 372.5K respectively. Which liquid has the highest vapour pressure at 323.K?
A. water B. toluene C. ethanol D. butan-2-ol E. none
37. The function of sulphur during the vulcanization of rubber is to: A. act as catalyst for the polymerization of rubber molecules B. convert rubber from thermosetting to thermoplastic polymer C. from chains which bind rubber molecules together D. break down rubber polymer molecule E. shorten the chain length of rubber polymer
38. Complete hydrogenation of ethyne yields
A. benzene B. methane C. ethane D. ethane E. propane
39. Mixing of aqueous solution of barium hydroxide and sodium tetraoxocarbonate (iv) yields a white precipitate of A. barium oxide B. sodium tetraoxocarbonate C. sodium oxide D. sodium hydroxide E. barium tetraoxocarbonate
40. An organic compound decolorized acidified KMnO_4 solution but failed to react with ammoniacal silver nitrate solution. The organic compound is likely to be
A. a carboxylic acid B. an alkane C. an alkene D. an alkyne E. an alkanone
41. Which of the following conducts electricity? A. sulphur B. graphite C. diamond D. red phosphorus E. yellow phosphorus
42. Which of the following compounds is not the correct product formed when the parent metal is heated in air? A. calcium oxide (CaO) B. sodium oxide (Na_2O) C. copper (ii) oxide (CuO) D. tri-iron tetroxide (Fe_3O_4) E. aluminium oxide (Al_2O_3)
43. When marble is heated to 1473 K, another whiter solid is obtained which reacts vigorously with water to give an alkaline solution. The solution contains
A. NaOH B. KOH C. $\text{Mg}(\text{OH})_2$ E. $\text{Ca}(\text{OH})_2$
44. Which of the following roles does sodium chloride play in soap preparation? It
A. react with glycerol B. purifies the soap C. accelerates the decomposition of fat and oil D. separates the soap from the glycerol E. converts the fat acid to its sodium salt
45. What is the least possible error in using a rule graduated in centimeters?
A. 0.1cm B. 0.5cm C. 1.0cm D. 2.0cm
46. Which of the following affects the period of simple pendulum? (i) mass of pendulum bob (ii) length of the pendulum (iii) acceleration due to gravity.
A. I, II and III B. II and III only C. I and III only D. I and II only
47. A boy sits in a train moving with a uniform speed on a straight track. If from his outstretched palm he gently tossed a coin vertically upwards, the coin will fall.
A. in front of his palm B. behind his palm C. beside of his palm D. into his palm
48. A machine required 1000J of work to raise a load of 500N through a vertical distance of 1.5m. Calculate the efficiency of the machine. A. 80% B. 75% C. 50% D. 33%
49. One of the most important application of bimetallic strip is found in the construction of
A. a thermostat B. an altimeter C. a thermocouple D. a hygrometer
50. At constant pressure, the density of a fixed gas is A. constant with temperature B. proportional to its volume C. inversely proportional to its temperature D. independent of its volume.

SOLUTION TO UNIPORT 2013/2014 POSTUTME SCREENING EXERCISE

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. D | 2. A | 3. B | 4. B | 5. C | 6. A | 7. C |
| 8. C | 9. B | 10. C | 11. D | 12. B | 13. B | 14. B |
| 15. B | 16. D | 17. C | 18. A | 19. B | 20. D | 21. B |
| 22. B | 23. | 24. B | 25. | | | |

UNIVERSITY OF PORTHARCOURT
2012/2013 POSTUTME SCREENING EXERCISE
FACULTY OF ENGINEERING TIME: 50MINS

1. Which of the following is/are water pollutants? A. fertilizers B. human wastes C. industrial wastes D. all of the above
2. Which of the following statements is correct of ΔH , the enthalpy of reaction? It is A. the heat change accompanying a chemical reaction B. negative for exothermic reactions C. positive for endothermic reactions D. all of the above
3. A saturated solution of potassium chloride was prepared at $x^{\circ}\text{C}$. if 30cm^3 of the solution required 22.5cm^3 of silver trioxonitrate (v) solution containing 34g in 100cm^3 solution for complete precipitation, calculate the solubility of potassium chloride at $x^{\circ}\text{C}$ in mol/dm^3 . [take $\text{KCl} = 74.5$, $\text{AgNO}_3 = 170$] A. 1.5mol/dm^3 B. 2.0mol/dm^3 C. 0.15mol/dm^3 D. 0.20mol/dm^3
4. The general methods of preparing salts include. i. neutralization ii. Precipitation iii. Double decomposition. A. I & ii B. iii & iv C. I, ii & iii D. I, ii, iii & iv
5. When a metal plate is placed in a solution containing its ions, some of the metallic ions in solution will take up electrons from the metal plate and deposit themselves as neutral atoms on the plate. Which option favours the reaction A. the electrode or metal plate becomes positively charged with respect to the solution/electrolyte B. the electrode or metal plate becomes neutral with respect to the solution/electrolyte C. the electrode or metal plate has more electrons with respect to the solution/electrolyte D. the electrode or metal plate becomes negatively charged with respect to the solution/electrolyte
6. How many isomers can be obtained from C_4H_{10} ? A. 0 B. 1 C. 2 D. 3
7. Who discovered the atom? A. Rutherford B. Dalton C. Neils Bohr D. Goldstein
8. 25cm^3 of a gas x contains z molecules at 15°C and 75mmHg . How many molecules will 25cm^3 of another gas y contain at the same temperature and pressure?
A. $2y$ B. $2z$ C. z D. y
9. The technique used for the separation of immiscible liquids is A. distillation
B. chromatography C. the use of separating funnel D. fractional distillation
10. Which of the following denotes an alpha particle? A. ${}^1_0\text{n}$ B. ${}^4_2\text{He}$ C. ${}^0_{-1}\text{e}$ D. ${}^9_4\text{Be}$
11. The shapes of CO_2 , H_2O and CH_4 respectively are: A. bent, linear and tetrahedral B. bent, tetrahedral and linear C. linear, bent and tetrahedral D. tetrahedral, linear and bent

Wives maximize their singleness and do not fret over whom they would be found by, how and when they would be found. They trust God always and strive to please him, knowing assuredly that all good and perfect gifts come from him. Wives cherish their period of preparation and are not in a hurry to get over with it, which makes them not to compromise in their standards, principles and virtues according to God's word. In fact, they are not in a hurry to jump out of their preparatory period for any man because they would only become incomplete and unfulfilled if they stopped half-way through their preparatory class. A woman may try to "impress" in order to seek the approval of some men, while wives are just impressive in all things. Not to get it wrong, wives may not be all that some men would want them to be, but most times they turn out to be what God wants of them to the man so that he also can become what God wants him to be. A woman may prefer you if your parents were dead, but a wife is groomed to handle the complicated patterns of relating with her in-laws. Wives are not rigid, rather they are malleable, trustworthy, loving and loveable, forgiving and forgivable, submissive and respected. They are almost always prepared for your ministry and are willing to stand by you at both good and bad times. Remember friends, except the woman is prepared she cannot be a wife. The period of sightedness is not just a period of waiting but more of the period of preparation and training for her man. If she's not prepared a wife, it may be difficult to make her one. Arise sisters, prepare yourselves, be wives and not just women, for it is only when a man finds a wife (and not a mere

woman) that he finds a good thing and obtains favour from the Lord. MARRIAGE IS ABOUT WIVES AND NOT WOMEN!

{Adapted from Peculiar Truth, 2002}

12. "...but most times they turn out to be what God wants of them to the man so that he also can become what God wants him to be" means that A. women are good advisers B. God makes wives for their husbands C. husbands are made gods by their wives D. wives help their husbands fit into God's plan
13. From the passage, when does a man find a good thing and obtain favour from the Lord? A. when he finds a girl B. when he loves his mother C. when he loves a woman D. when he finds a wife
14. Chike must have passed the house five times yet, he was still not sure. Was this the house he had visited so often in the past? The house he used to visit was a bungalow. This house too, was a bungalow. The old house was situated between two storied buildings; this one too, was so situated. But in spite of this feeling of certainty, Chike had a vague suspicion that the house he had passed so often that day might be the wrong house after all. Could an absence of two years have blurred his memory so badly? After a few moments' hesitation, Chike began to move towards the house and then stopped, as if held back by an invisible hand. His attention had been attracted by a girl of about twenty-four, who was tripping along the pavement to his right. He turned and advanced towards her, and was about to call her by what he felt was her name when he discovered that he had made a mistake in respect of her identity. Just as Chike turned from the girl to continue his quest, he heard voices shouting, 'thief! Thief!' and saw a crowd materialize in seconds. At the head of this justice-impelled rabble, was a ludicrously fat woman who, inspite of her size, bounced along with the agility of a prize athlete. This barrel bellowed loudest and clapped her hands vigorously to punctuate her outburst. And as she did so, the surplus flesh on her podgy arms quivered. 'they are all the same,' this woman screamed, addressing nobody in particular. 'They dress gorgeously, but underneath they are rogues'. Again, she clapped her hands and again, there was a quivering of loose flesh. Chike was so busily occupied with watching the antics of this woman that he did not notice at first that the object of her venom is the girl that he had seen earlier. It was this girl that was now surrounded by the crowd, with many people groaning, sighing and hissing in unison. Chike relaxed and prepared to watch the drama unfolding before him.

The expression 'justice-impelled' in the passage refers to the desire of the crowd to

- A. See that the case was taken before a judge B. take the thief to the police station
 - C. try the thief immediately D. see that there was fair play
15. Standard English refers to the authoritative and correct usage of the language, the medium of expression for government and education. Its opposite is a dialectal variant of the language, that is, accepted and recognized words, expressions and structures peculiar to a smaller group of language users who are generally set apart from standard usage by cultural group or geographical region. For example, Nigerian, American, Irish and British English differ from one another in many respects and each is identifiable, yet in every case the standard variety approaches a single and hypothetical classification known as international English. As one moves towards informality and away from the observance of strict rules, emphasis falls on the differences between dialects. In addition to American English being distinguishable from British English, it is also true that British English is not uniform within the United Kingdom. The level of formality is determined by education and aspiration, while dialects vary from region to region (*Jamb 1995*)

One characteristic of a dialect as mentioned in the passage is A. informality B. possession of various forms C. distinction from British English D. restricted area of usage

16. Identify the sentence with the wrong punctuation mark. A. he doesn't know the answer, does he? B. she has lots of cats at home C. Joe's sister is in the Himalayas D. the problem with the business is its location
17. One fact we have to comprehend is that in our unconscious mind, we cannot distinguish between a wish and a deed. We are all aware of some of our illogical dreams in which two completely opposite statements can exist side by side – very acceptable in our dreams but unthinkable and illogical in our waking state. Just as our unconscious mind cannot differentiate between the wish to kill somebody in anger and the act of having done so, the young child is unable to make this distinction. The child who angrily wishes his mother to drop dead for not having gratified his need will be traumatized greatly by the actual death of his mother – even if this event is not linked closely in time with his destructive wishes. He will always take part of or the whole of the blame for the loss of his mother. He will say to himself, rarely to others, 'I did it. I am responsible. I was bad, therefore mummy left me'. It is well to remember that the child will react in the same manner if he loses a parent by divorce, separation or desertion. Death is often seen by the child as an impeachment and therefore has little distinction from a divorce in which he may have an opportunity to see a parent again.

The passage emphasizes A. a child growing up in ignorance B. our unconscious mind C. a child's inability to distinguish between dream and reality D. illogical dreams

18. is the owner of this pair of scissors? A. who B. whom C. which D. whose
19. This..... be Asawo's handwriting; I know his handwriting well enough
A. may B. will C. ought D. can't
20. Amina used to be *hand-in-glove* with her boyfriend. A. cheat on her boyfriend B. admire her boyfriend C. fight her boyfriend D. wear her boyfriend's glove E. be in a very close relationship with her boyfriend
21. This masquerade appears *once in a blue moon*. This means that the masquerade appears.
A. on very rare occasions B. when the moon is blue C. whenever a special request is made D. once a month
22. In the sentence "John considered himself *successful*", the italicized word is correctly classified as A. indirect object B. predicate nominative C. direct object D. object complement E. predicate adjective
23. In the sentence "My father, who is standing over there by the wall, is a *very* good dancer", the italicized word is an example of A. an adverb B. a coordinating conjunction C. an adjective D. an indefinite pronoun
24. The sentence "the student didn't know the rules until they were explained to him", contains
A. co-ordinating conjunction B. subordinating conjunction C. conjunctive adverb D. modal verb
25. The lengths of the parallel sides of a trapezium are 9cm and 12cm. if the area of the trapezium is 105cm^2 , find the perpendicular distance between the parallel sides.
A. 5cm B. 7cm C. 10cm D. 15cm
26. Seven students are to have a jolly good time together. If they are to be seated, find the number of ways they are to be chosen. A. 5040 B. 6040 C. 4050 D. 4060
27. A rope of length 11cm is folded to form a sector of a circle of radius 3.5m, on a school playing field. What is the size of the angle of the sector, correct to the nearest degree.
A. 33 B. 40 C. 90 D. 180
28. A man obtained N48,000.00 in exchange for \$600.00. what is the equivalent of one dollar to the naira in the exchange? A. N0.8 B. N1.25 C. N80.00 D. N125.00
29. Find the length of an arc of a circle having radius 66cm and subtends an angle of 60° at the circle centre. A. 40cm B. 55cm C. 90cm D. 69cm
30. x varies directly as the square of y and inversely as p, when y = 5, x = 8 and p = 10. Find the value of x when y = 3 and p = $\frac{1}{3}$. A. 28.8 B. 86.4 C. 82.8 D. 68.4

31. Express the product of 0.00267 and 0.00486 in standard form.
A. 1.2 B. 1.3×10^{-6} C. 1.30 D. 1.30×10^{-5} E. 1.30×10^5
32. Let the probability function on set S, where $S = [a_1, a_2, a_3, a_4]$. Find $P(a_1)$ if $P(a_2) = \frac{1}{3}$, $P(a_3) = \frac{1}{6}$ and $P(a_4) = \frac{1}{5}$. A. $\frac{7}{10}$ B. $\frac{2}{3}$ C. $\frac{1}{3}$ D. $\frac{3}{10}$
33. In an election there were four candidates, $\frac{5}{6}$ of the electors voted for the winner. The runner-up received $\frac{4}{7}$ of the remaining votes. The third candidates received twice the votes of the fourth candidates. If the winner received 3021 votes more than the runner-up, how many electors voted? A. 126,882 B. 63,441 C. 18,126 D. 35,245
34. During two years in a school, $\frac{7}{8}$ of the students had malaria, $\frac{3}{4}$ had catarrh and $\frac{5}{6}$ had neither. What fraction of the school had both malaria and catarrh?
A. $1\frac{1}{2}$ B. $1\frac{11}{24}$ C. $1\frac{1}{24}$ D. $\frac{1}{2}$
35. $\frac{3}{4}$ of a class of 32 students study English and $\frac{3}{8}$ study Mathematics. Every student studies at least one of these subjects. What fraction of the class studies Mathematics but not English?
A. $\frac{1}{4}$ B. $\frac{5}{8}$ C. $\frac{3}{8}$ D. $\frac{8}{9}$
36. Water flows through a 3cm diameter pipe at the rate of 3metre/second. How many cm^3 of water flow through the pipe in one second.
A. $450\text{cm}^3/\text{s}$ B. $900\text{cm}^3/\text{s}$ C. $225\text{cm}^3/\text{s}$ D. $2121\text{cm}^3/\text{s}$
37. The product of two numbers is 12. The sum of the larger number and twice the smaller is 11. Find the two numbers. A. 4, 3 B. 8, $2\frac{1}{2}$ C. 3, $1\frac{1}{2}$ D. $1\frac{1}{2}$, 4
38. Divide 1324_8 by 20201_3 and leave your answer in binary numeral.
A. 101_2 B. 011_2 C. 010_2 D. 100_2
39. A city B is 205km from a village C on a bearing 035° . How far is B east of C?
A. 205km B. 146km C. 117.6km D. 91.7km
40. If two graphs $y = px^2 + q$ and $y = 3x^2 - 1$ intersect at $x = 2$, find the value of p in terms of q.
A. $\frac{11-q}{4}$ B. $\frac{q-12}{11}$ C. $\frac{12-q}{4}$ D. $\frac{11+q}{4}$
41. Calculate the force acting on an electron of charge $1.6 \times 10^{-19}\text{C}$ placed in an electric field of intensity 10^8V/m . A. $1.6 \times 10^{-11}\text{N}$ B. $1.6 \times 10^{-14}\text{N}$ C. $1.6 \times 10^{-16}\text{N}$ D. $1.0 \times 10^{-16}\text{N}$
42. A body of mass 400g is whirled in a vertical circle of radius 2m at a steady rate of 2rev/s. find its centripetal force. A. 160N B. 102N C. 320N D. 252N
43. In A.C circuit theory, the root mean square (r.m.s) current, I_{rms} and the peak (maximum) current, I_0 are related by. A. $I_0 = I_{\text{rms}}/\sigma\pi\alpha v > 2$ B. $I_{\text{rms}} = I_0/\sigma\pi\alpha v > 2$ C. $I_{\text{rms}} = I/(I_0/\sigma\pi\alpha v > 2)$ D. $I_0 = I/(I_{\text{rms}}/\sigma\pi\alpha v > 2)$
44. If the volume of a fixed mass of a gas is kept constant, the pressure of the gas ...
A. is inversely proportional to its centigrade temperature B. is inversely proportional to its absolute temperature C. is directly proportional to its centigrade temperature D. is directly proportional to its absolute temperature
45. Find the reactance of a capacitor $500/\pi\text{Hz}$ frequency if in a circuit a capacitor has a reactance = 5F. A. 200Ω B. 200.5Ω C. 80.7Ω D. 190.1Ω
46. A 500kg car which was initially at rest traveled with an acceleration of 5m/s^2 , its kinetic energy after 4s was A. 10^5J B. 10^4J C. 10^3J D. 10^2J
47. Evaporation is affected by all EXCEPT A. wind and air dryness B. the liquid's nature C. temperature D. the freeness of the liquid
48. To reduce heat loss in a transformer, A. the number of turns should be increased B. the soft iron core should be laminated C. the area of the coils in the soft iron core should be increased D. all of the above
49. Calculate the power in operation when a force of 10N moves a distance of 5m at a rate of 2secs.
A. 25W B. 50W C. 5W D. 100W
50. The process of magnetizing an object of magnetic material simply by bringing a magnet near is called A. electromagnetic induction B. ionic magnetization C. electrolytic magnetization D. induced magnetism

SOLUTION TO UNIPORT 2012/2013 POSTUTME SCREENING EXERCISE

1. D	2. D	3. A	4. D	5. A	6. C	7. B
8. C	9. C	10. B	11. C	12. D	13. D	14. D
15. D	16. B	17. C	18. A	19. D	20. E	21. A
22. C	23. A	24. B	25. C	26. A	27. D	28. C
29. D	30. B	31. D	32. D	33. A	34. B	35. A
36. D	37. A	38. D	39. C	40. A	41. A	42. D
43. B	44. D	45. A	46. A	47. D	48. B	49. A
50. A						

UNIVERSITY OF PORTHARCOURT 2011/2012 POSTUTME SCREENING EXERCISE TIME: 50MINS

PAPER 1

1. A compound contains 40% Carbon, 6.7% Hydrogen and 53.3% Oxygen. If the molar mass of the compound is 180, find the molecular formula. A. CH_2O B. $\text{C}_6\text{H}_6\text{O}_3$ C. $\text{C}_6\text{H}_{12}\text{O}_6$ D. $\text{C}_6\text{H}_6\text{O}_3$
2. Aluminium hydroxide is used in the dyeing industry as a A. salt B. dye C. mordant D. dispersant
3. An element that occurs in the free state is A. Pb B. K C. Al D. Ar E. C
4. $\text{C}_2\text{H}_4(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{C}_2\text{H}_6(\text{g})$ $\Delta H = -13$
The reaction represented above is A. exothermic B. spontaneous C. endothermic D. in equilibrium
5. Calculate the mass of chlorine gas which occupies a volume of 1.12dm^3 at S.T.P. A. 15.50g B. 7.10g C. 3.55g D. 1.80g [$\text{Cl} = 35.5$, $\text{M.V.G} = 22.4\text{dm}^3$]
6. Calculate the number of moles of HCl present in 20cm^3 of a 0.75M solution of the acid. A. 0.015mole B. 0.038mole C. 3.800mole D. 1.500mole
7. Chlorine consisting of 2 isotopes of mass numbers 35 and 37 in the ratio 3:1 has an atomic mass of 35.5. calculate the relative abundance of the isotope of number 3?
A. 20 B. 25 C. 60 D. 75
8. How long will it take to deposit 0.08g of copper from CuCl_2 solution by passing a current of 0.5A. A. 6mins B. 8mins C. 24mins D. 48mins [$\text{Cu} = 64$, $F = 96500\text{C/mol}$]
9. How much faster does a Helium atom travel than a nitrogen molecule at the same temperature? A. 1.68 B. 1.70 C. 2.50 D. 2.65 E. 2.90
10. If a solution contains 4.9g of tetraoxosulphate (VI) acid, calculate the amount of copper (II) oxide that will react with it. A. 0.8g B. 4.0g C. 40.0g D. 80.0g
11. $\text{MnO}_{2(\text{s})} + x\text{HCl}_{(\text{aq})} \rightarrow \text{MnCl}_{2(\text{aq})} + y\text{H}_2\text{O}_{(\text{l})} + z\text{Cl}_{2(\text{g})}$ In the equation above, what are the values of x, y and z respectively? A. 2, 1, 2 B. 4, 1, 2 C. 4, 2, 1 D. 1, 2, 1
12. Tartaric acid is used industrially to A. make baking powder B. make fruit juices C. remove rust D. dry substances
13. The carbon atoms in ethane are hybridized. A. sp^4 B. sp^3 C. sp^2 D. sp E. none
14. The mass of silver deposited when a current of 10A is passed through a solution of silver salt for 4830s is A. 108.0g B. 54.0g C. 27.0g D. 13.5g [$\text{Ag} = 108$, $F = 96500\text{C/mol}$]
15. The oxidation state of oxygen in tetraoxosulphate (VI) acid is A. -4 B. -2 C. +4 D. -8
16. The reaction that takes place in Daniel cell is A. direct combination B. redox C. double decomposition D. neutralization
17. The salt formed from a strong acid and a strong base is A. complex B. neutral C. basic D. acidic

18. The type of reaction that is peculiar to benzene is A. addition B. hydrolysis C. polymerization D. substitution
19. Under normal condition of ionic addition, what will be the major product of the addition of HBr to 2-methyl-2-butene. A. 2-Bromo-3-methylbutane B. 1-Bromo-3-methylbutane C. 2-methyl-1-bromobutane D. 2-bromo-2-methylbutane
20. What is the percentage by mass of oxygen in $\text{Al}_2(\text{SO}_4)_3 \cdot 2\text{H}_2\text{O}$?
A. 14.29% B. 25.39% C. 50.79% D. 59.25%
21. What volume of $0.5\text{mol/dm}^3 \text{H}_2\text{SO}_4$ will exactly neutralize 20cm^3 of $0.1\text{mol/dm}^3 \text{NaOH}$ solution?
A. 2.0cm^3 B. 5.0cm^3 C. 6.8cm^3 D. 8.3cm^3
22. What volume of gas is evolved at s.t.p if 2g of calcium trioxocarbonate (IV) is added to a solution of hydrochloric acid? A. 112cm^3 B. 224cm^3 C. 448cm^3 D. 2240cm^3
[Ca = 40, C = 12, O = 16, Cl = 35.5, H = 1, molar volume of gas at s.t.p = 22.4dm^3]
23. Which experiment led to the measurement of the charge on an electron? A. mass spectrometric experiment B. oil-drop experiment C. scattering a particle D. discharge tube experiment
24. Which of the following acts as both a reducing and an oxidizing agent?
A. H_2 B. SO_2 C. H_2S D. CO_2
25. Which of the following is not a unit of temperature? A. $^\circ\text{C}$ B. $^\circ\text{F}$ C. K D. $^\circ\text{K}$ E. None of the above
26. 2 spheres of masses 5kg and 10kg are 0.3m apart. Calculate the force of attraction between them [$G = 6.67 \times 10^{-11}\text{Nm}^2\text{Kg}^{-2}$] A. $3.50 \times 10^{-10}\text{N}$ B. $3.71 \times 10^{-8}\text{N}$ C. $3.57 \times 10^{-2}\text{N}$ D. $4.00 \times 10^{-2}\text{N}$
27. A 2000W heater is used to heat a metal object of mass 5kg initially at 10°C . If a temperature rise of 30°C is obtained after 10min, the heat capacity of the material is
A. $1.2 \times 10^4\text{J/}^\circ\text{C}$ B. $6.0 \times 10^4\text{J/}^\circ\text{C}$ C. $8.0 \times 10^4\text{J/}^\circ\text{C}$ D. $40 \times 10^4\text{J/}^\circ\text{C}$
28. A boy drags a bag of rice along a smooth horizontal floor with a force of 2N applied at angle of 60° to the floor. The work done after a distance of 3m is A. 3J B. 4J C. 5J D. 6J
29. A car of mass 1500kg goes round a circular curve of radius 50m at a speed of 40m/s. the magnitude of the centripetal force on the car is A. $4.8 \times 10^4\text{N}$ B. $4.8 \times 10^3\text{N}$ C. $1.2 \times 10^3\text{N}$ D. $1.2 \times 10^4\text{N}$
30. A converging lens of 15cm focal-length forms an image of an object placed 9cm from it. What is its position and magnification? A. -6cm and $\frac{3}{2}$ B. -6cm and $\frac{1}{3}$ C. -6cm and $-\frac{2}{3}$ D. -6cm and $\frac{2}{3}$ E. $\frac{2}{3}$ and -6cm
31. A gramophone record takes 5s to reach its constant angular velocity of $4\pi\text{rad/s}$ from the rest. Find its constant angular acceleration. A. $0.4\pi\text{rad/s}^2$ B. $0.8\pi\text{rad/s}^2$ C. $1.3\pi\text{rad/s}^2$ D. $20.0\pi\text{rad/s}^2$
32. A machine has a velocity ratio of 4. If it requires 800N to overcome a load of 1600N, what is the efficiency of the machine? A. 60% B. 50% C. 40% D. 40% E. 2%
33. A piece of iron weighs 250N in air and 200N in a liquid of density 1000kg/m^3 . The volume of the iron is. A. $5.0 \times 10^{-3}\text{m}^3$ B. $4.5 \times 10^{-3}\text{m}^3$ C. $2.5 \times 10^{-3}\text{m}^3$ D. $2.0 \times 10^{-3}\text{m}^3$
34. A ray of light is incident on an equilateral triangular glass prism of refractive index $\frac{3}{2}$. Calculate the angle through which the ray is minimally deviated in the prism.
A. 37.2° B. 48.6° C. 30.0° D. 42.0°
35. A reservoir 500m deep is filled with a fluid of density 850kg/m^3 . If the atmospheric pressure is $1.05 \times 10^5\text{N/m}^2$, the pressure at the bottom of the reservoir is [$g = 10\text{m/s}^2$]
A. $4.72 \times 10^6\text{N/m}^2$ B. $4.36 \times 10^6\text{N/m}^2$ C. $4.28 \times 10^6\text{N/m}^2$ D. $4.25 \times 10^6\text{N/m}^2$
36. A string of length 4m is extended by 0.02m when a load of 0.4kg is suspended at its end. What will be the length of the string when the applied force is 15N? [$g = 10\text{m/s}^2$]
A. 4.05m B. 4.08m C. 5.05m D. 6.08m
37. A tuning fork of frequency 340Hz is vibrated just above cylindrical tube of height 1.2m. if water is slowly poured into the tube, at what maximum height will resonance occur? [speed of sound in air = 340m/s] A. 0.45m B. 0.95m C. 0.60m D. 0.50m

38. A wire of 5Ω resistance is drawn out so that its new length is twice the original length. If the receptivity of the wire remains the same and the cross sectional area is halved, the new resistance is A. 20Ω B. 5Ω C. 40Ω D. 10Ω
39. An electron of charge $1.6 \times 10^{-19}\text{C}$ is accelerated between 2 metal plates. If the kinetic energy of the electron is $4.8 \times 10^{-17}\text{J}$, the potential difference between the plates is A. 300V B. 30V C. 400V D. 40V
40. An object is placed 20cm in front of a concave mirror whose focal length is 10cm . where and how is the image formed? A. centre of curvature and inverted B. 20cm behind the mirror and inverted C. 20cm in front of the mirror and upright D. 20cm in front of the mirror and virtual E. centre of curvature and virtual
41. Caesium has a work function of $3 \times 10^{-19}\text{J}$. the maximum energy of liberated electrons when it is illuminated by light of frequency $6.7 \times 10^{14}\text{Hz}$ is A. $1.42 \times 10^{-19}\text{J}$ B. $3.00 \times 10^{-19}\text{J}$ C. $4.42 \times 10^{-19}\text{J}$ D. $7.42 \times 10^{-19}\text{J}$ [$h = 6.63 \times 10^{-34}$]
42. Given that Young's modulus for aluminium is $7.0 \times 10^{10}\text{N/m}^2$ and density is $2.7 \times 10^3\text{kg/m}^3$. Find the speed of sound produced if a solid bar is struck at one end with a hammer. A. $3.6 \times 10^3\text{m/s}$ B. $5.1 \times 10^3\text{m/s}$ C. $2.8 \times 10^3\text{m/s}$ D. $4.2 \times 10^3\text{m/s}$
43. If $1.2 \times 10^6\text{J}$ of heat energy is given off in 1sec from a vessel maintained at a temperature gradient of 30K/m , the surface area of the vessel is A. $1.0 \times 10^3\text{m}^2$ B. $1.0 \times 10^2\text{m}^2$ C. $9.0 \times 10^4\text{m}^2$ D. $9.0 \times 10^2\text{m}^2$ [thermal conductivity of the vessel = 400W/m/K]
44. Light ray is incident on air-glass interface at angle of 70° to the interface. Given that the refractive index of glass is 1.45, what refraction angle will the ray make with the normal? A. 20.41° B. 20.71° C. 20.17° D. 18.97° E. 25.00°
45. On a rainy day of 20°C , the length of a steel truck is 20m . what will be its length on a hot day with temperature of 40°C . A. 20.009m B. 20.002m C. 20.013m D. 20.004m [coefficient of linear expansion of steel = $11 \times 10^{-6}\text{K}^{-1}$]
46. One of the following phenomena must occur whenever white light is dispersed. A. interferes B. polarization C. reflection D. diffraction
47. The count rate of a radioactive material is 800count/min . if the half-life of the material is 4 days, what would the count rate be 16 days later? A. 50count/min B. 25count/min C. 200count/min D. 100count/min
48. The force on a charge moving with velocity v in a magnetic field B is half of the maximum force when the angle between v and B is A. 0° B. 90° C. 45° D. 30°
49. The vertical displacement experienced by a vibrating body during wave motion is called A. period B. a cycle C. amplitude D. displacement E. distance
50. Water in an open vessel boils at a lower temperature when heated at the mountain top because A. pressure is increased B. pressure is reduced C. the rays of sun assists D. the saturated vapour pressure is eliminated E. none of the above

PAPER 2

Choose the option opposite in meaning to the word or phrase underlined

1. "A novel is an embellished falsehood", said the teacher. A. enriched B. exaggerated C. adorned D. obliterated
2. curiously, he escaped unhurt. A. interestingly B. unsurprisingly C. annoyingly D. unusually
3. His comment was not printable. A. punishable B. suitable C. offensive D. unfair
4. I have had enough of your impudence. A. politeness B. incivility C. arrogance D. boldness
5. Sola was apathetic about his work. A. indifferent B. enthusiastic C. halfhearted D. bored
6. The compulsory leave made her feel listless. A. restless B. great C. fat D. energetic
7. We were collectively responsible for keeping the machine in good shape
A. jointly B. severally C. dutifully D. socially

Select the option that best explains the information conveyed in the sentence

8. Emeka was always conspicuous by his absence when it came to attending lectures
 A. that Emeka always attended lectures made him conspicuous at school B. Emeka's absence from lectures was all too glaring C. it was obvious that Emeka was not present at lectures
 D. what made Emeka conspicuous was his occasional absence from school
9. He was injured by friendly fire A. he was wounded in a war by a friend who fired a shot mistakenly B. he was injured by a gunshot that was fired by a friend C. he was hit in a war by a weapon fired by his own side D. it was a war and he was hit by an unknown soldier fired by an unknown soldier

Choose the option nearest in meaning to the word(s) or phrase underlined

10. 'You're impossible', she said to me A. unreasonable B. unforgiving C. daft D. strong
11. Everyone was talking about the camaraderie's in the club.
 A. companionship B. quarrelsomeness C. commissioning D. gentle
12. I was being directed by the sixth sense all through the ordeal. A. intelligent B. judgement
 C. intuition D. imagination
13. Tade has an aquiline nose. A. large and curved nose B. a wide and straight nose C. big and pointed nose D. a small and attracted nose
14. The man's wife has her misgivings about him. A. fears for B. distrust of C. hatred for
 D. love for
15. The rest of the boys admire Olu's verve. A. energy B. handsomeness C. work D. intelligence
16. The tax rebate was a concession given to the civil servants A. an incentive B. an allowance
 C. work D. intelligence
17. Violent crime is only one of the maladies affecting modern society. A. damages B. illnesses
 C. wounds D. crisis

Choose the option that best completes the gap

18. Birds whistle while bears A. giber B. purr C. growl D. bellow
19. I am not surprised that Sani had a road accident, he was always driving too fast, ...?
 A. isn't he B. wasn't he C. didn't he D. did he
20. I will pay my dept A. instalmentally B. by instalmentally C. in instalments D. in instalmentally
21. Oduma had enough money to pay the fines so he went to jail for distributing libelous ...
 A. no / brochures B. not / circulars C. no / publications D. not / pamphlets
22. She refrained smoking. A. at B. off C. from D. in
23. The litigants finally opted settlement. A. by an out of court B. in an out of court C. to an out of court D. for an out of court
24. The scholar examined of the subjects. A. three apart analysis B. apart three analysis
 C. a three part analysis D. three part analysis
25. You should anticipate and prepare for changes in your office. A. that has taken place
 B. which takes place C. taking place D. that would take place

MATHEMATICS

26. Let A be an event with probability $P(A) = 0.14$. find the probability of the complement of event A
 A. 0.14 B. -0.86 C. 1.0 D. 0.86
27. A box contains 5 blue balls, 4 green balls, if a ball is chosen at random, what is the probability that the ball is either blue or red? A. $\frac{3}{12}$ B. $\frac{5}{12}$ C. $\frac{8}{12}$ D. $\frac{15}{12}$
28. Solve the simultaneous linear equations: $2x + 5y = 1$; $7x + 4y = 2$
29. Solve the system of equation: $2^{x+y} = 32$; $3^{3y-x} = 27$. A. (3, 2) B. (-3, 2) C. (3, 2) D. (-3, 2)
 E. (3, 2)

30. The chances of three independent events X, Y, Z occurring are $\frac{1}{2}$, $\frac{2}{3}$, $\frac{1}{4}$ respectively. What are the chances of Y and Z only occurring? A. $\frac{1}{8}$ B. $\frac{1}{24}$ C. $\frac{1}{12}$ D. $\frac{1}{4}$

SOLUTION TO UNIPOINT 2011/2012 POSTUTME SCREENING EXERCISE

PAPER 1

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. C | 3. D | 4. A | 5. C | 6. A | 7. B |
| 8. B | 9. | 10. B | 11. C | 12. A | 13. B | 14. B |
| 15. B | 16. B | 17. B | 18. D | 19. D | 20. D | 21. A |
| 22. C | 23. B | 24. B | 25. D | 26. B | 27. A | 28. A |
| 29. A | 30. D | 31. B | 32. B | 33. A | 34. A | 35. |
| 36. | 37. | 38. | 39. A | 40. | 41. C | 42. |
| 43. | 44. | 45. | 46. B | 47. | 48. C | 49. C |
| 50. A | | | | | | |

PAPER 2

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. | 2. | 3. B | 4. A | 5. B | 6. B | 7. A |
| 8. C | 9. C | 10. A | 11. A | 12. B | 13. D | 14. B |
| 15. A | 16. C | 17. D | 18. B | 19. A | 20. B | 21. B |
| 22. C | 23. D | 24. C | 25. D | | | |

MATHEMATICS

26. D 27. 28. 29. A 30.

UNIVERSITY OF PORTHARCOURT 2010/2011 POSTUTME SCREENING EXERCISE TIME: 50MINS

INSTRUCTION: Shade all answer in computer sheet

MATHEMATICS

- Find the slope of the curve $y = 2x^2 + 5x - 3$ at (1, 4). A. 4 B. 0 C. 7 D. 9
- Determine the maximum value of $y = 3x^2 - x^3$. A. 0 B. 2 C. 4 D. 6
- By how much will the mean of 30, 56, 31, 55, 43 and 44 less than the median.
A. 0.75 B. 0.50 C. 0.33 D. 0.17
- The range of 4, 3, 11, 9, 6, 15, 19, 23, 27, 24, 21 and 61 is A. 16 B. 21 C. 23 D. 24
- The mean of the numbers 3, 6, 4, x and 7 is 5, find the standard deviation.
A. $\sqrt{2}$ B. $\sqrt{3}$ C. 2 D. 3
- Find the remainder when $3x^3 + 5x^2 - 11x + 4$ is divided by $x + 3$. A. 4 B. 1 C. -1 D. -4
- What are the integral values of x which satisfy the inequality $3 - 2x \leq 5$?
A. -6 B. -4 C. -1 D. -4
- Find the derivative of $(2x + 3x)(1 - x)$ with respect to x A. $6x - 1$ B. $1 - 6x$ C. 6 D. -3
- Find the derivative of the function $y = 2x^2(2x - 1)$ at the point $x = -1$. A. -6 B. -4 C. 16 D. 18
- Find the mean deviation of 1, 2, 3 and 4. A. 1.0 B. 1.5 C. 2.0 D. 2.5
- Find the value of t if the standard deviation of 2t, 3t, 5t and 6t is $\sqrt{2}$. A. 1 B. 2 C. 3 D. 4
- In how many ways can 6 coloured chalks be arranged if 2 are of the same colour?
A. 60 B. 120 C. 240 D. 360

13. A final examination requires that a student answer any 4 out of 6 questions. In how many ways can this be done? A. 15 B. 20 C. 30 D. 45
14. If the mean of five consecutive integers is 30, find the largest of the numbers
A. 28 B. 30 C. 45 D. 40
15. A bag contains 5 black, 4 white and x red marbles. If the probability of picking a red marble is $\frac{2}{5}$. Find the value of x. A. 8 B. 10 C. 4 D. 6
16. For what value s of n is ${}^{n-1}C_3 = 4({}^nC_3)$? A. 6 B. 5 C. 4 D. 3
17. Find the roots of $x^3 - 2x^2 - 5x + 6 = 0$. A. 1, 2, -3 B. -1, 2, 3 C. -1, 2, 3 D. 1, -2, 3
18. Find the value of k if the expression $kx^3 + x^2 - 5x - 2$ leaves a remainder 2 when it is divided by $2x + 1$. A. 10 B. 0 C. -10 D. -8
19. If $y = x^2 - x - 12$. Find the range of values of x for which $x \geq 0$. A. $x < -3$ or $x > -1$ B. $x \leq -3$ or $x \geq 4$ C. $-3 < x \leq 4$ D. $-3 \leq x \leq 4$
20. How many terms of the series 3, -6, +12, -24, : are needed to make a total of $1 - 2^8$.
A. 12 B. 10 C. 9 D. 8

PHYSICS

21. The wavelength of the overtone of a note in a closed pipe of length 33cm is
A. 44cm B. 33cm C. 22cm D. 17cm
22. Non-luminous objects can be seen because they A. are polished B. are near C. reflect light D. emit light
23. The correct unit of energy density is A. $\text{kgm}^{-3}\text{s}^{-2}$ B. $\text{kgm}^{-1}\text{s}^{-2}$ C. $\text{kgm}^{-3}\text{s}^{-2}$ D. $\text{kgm}^2\text{s}^{-2}$
24. The motion of smoke particles from a chimney is typical of A. oscillatory motion B. rotational motion C. Circular motion D. random motion
25. One of the properties of gamma rays is that they are A. negatively charged B. massive C. neutral D. positively charged
26. The process whereby the molecules of different substances move randomly is called A. surface tension B. diffusion C. capillarity D. osmosis
27. The process whereby a liquid turns spontaneously into vapour is called
A. evaporation B. relegation C. boiling D. sublimation
28. The velocity of sound in air will be doubled if its absolute temperature is A. doubled B. halved C. constant D. quadrupled
29. A thin converging lens has a power of 4.0 diopters. determine its focal length
A. 0.25m B. 0.03m C. 5.00m D. 2.50m
30. An electric device is rated 2000W, 250V, the correct fuse rating of the device is A. 8A B. 9A C. 7A D. 6A
31. Satellite communication network makes use of A. infrared rays B. sound wave C. visible light D. radio wave
32. If two inductors of inductance 3H and 6H are arranged in series, the total inductance is
A. 18.0H B. 9.0H C. 2.0H D. 0.5H
33. The current in a reverse biased junction is due to A. electrons B. majority carriers C. holes D. minority carriers
34. In an A.C circuit that contains only a capacitor, the voltage lags behind the current by
A. 0° B. 180° C. 60° D. 30°
35. The ray which causes gas molecules to glow is known as A. molecular ray B. gamma ray C. anode ray D. cathode ray
36. The charge carriers in gases are A. ions only B. electrons and holes C. electrons only D. electrons and ions
37. Which of the following materials is a conductor? A. plastic B. sodium C. wax D. glass
38. The instrument used for securing a large number of similar charges by induction is called
A. capacitor B. electrophorus C. electroscope D. proof-plane
39. The pitch of a sound note depends on A. timbre B. harmonics C. quality D. frequency

40. In which of the following material media would sound travel fastest A. water B. oil C. metal D. gas

CHEMISTRY

41. Maximum number of electron is found in one of these.... A. 4s B. 4p C. 4d D. 4f
42. The shape of the s-orbital is A. spherical B. elliptical C. spiral D. circular
43. A carcinogenic substance is A. asbestos dust B. sawdust C. nitrogen (II) oxide D. carbon (II) oxide
44. In the electrolysis of brine, the anode is A. platinum B. copper C. zinc D. carbon
45. Which of the following hydrogen halides has the highest entropy value?
A. HF B. HCl C. HBr D. HI
46. The allotrope of carbon used in the decolonization of Niger is A. graphite B. soot C. charcoal D. lampblack
47. Sulphur (IV) oxide bleaches by A. reduction B. oxidation C. hydration D. absorption
48. Aluminum hydroxide is used in the dyeing industry as a A. salt B. dye C. mordant D. dispersant
49. An isomer of C_5H_{12} is A. butane B. 2-methylbutane C. 2-ethylbutane D. 2-ethylbutene
50. Vulcanization involves the removal of A. a monomer B. the single bond C. the double bond D. a polymer
51. Phenolphthalein in acidic solution is A. red B. orange C. colourless D. yellow
52. When iron is exposed to moisture and it rusts, the value of ΔG for the reaction is
A. neutral B. zero C. positive D. negative
53. A substance that is used as a ripening agent for fruits is A. ethane B. propane C. methane D. butane
54. The shape of the hydrocarbon compound CH_4 is A. square planar B. planar C. linear D. tetrahedral
55. Sugar is separated from its impurities by A. precipitation B. crystallization C. distillation D. evaporation
56. The component of an atom that contributes least to the mass is the A. proton B. nucleus C. neutron D. electron
57. An element will readily form an electrovalent compound if its electron configuration is
A. 2, 8, 1 B. 2, 8, 4 C. 2, 8, 8, 12 D. 2, 8, 5
58. The most suitable metal that can be used as a lightning conductor is A. silver B. copper C. iron D. aluminum
59. The most abundant element on the earth's crust is A. nitrogen B. hydrogen C. oxygen D. fluorine
60. Metalloids are also referred to as A. semi-metals B. metals C. colloids D. non-metals
61. The ores that can be concentrated by flotation are: A. nitride ores B. sulphide ores C. oxide ores D. chloride ores

From the list of words, choose the one that best completes each sentence from 81 to 95

62. You are driving fast for my liking. A. too B. very C. pretty D. fairly
63. You have given me one oranges many A. very B. so C. too D. more
64. The upholstery work doesn't go the colour of the car. A. after B. by C. with D. in
65. I became depressed hearing the news. A. at B. with C. as D. on
66. He was punished for failing his duty as a prefect of the school. A. on B. about C. with D. in
67. Good discipline was instructed the success achieved by the college.
A. for B. of C. in D. with
68. It was quite dark in the room we couldn't see A. so B. because C. through D. yet

SOLUTION TO UNIPORT 2010/2011 POSTUTME SCREENING EXERCISE

MATHEMATICS

- | | | | | | | |
|------|------|------------|----|----|------|------|
| 1. D | 2. C | 3. B | 4. | 5. | 6. B | 7. B |
| 8. | 9. | 10. A | | | | |

PHYSICS

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 21. B | 22. C | 23. A | 24. D | 25. B | 26. B | 27. A |
| 28. D | 29. A | 30. A | 31. B | 32. C | 33. | 34 |
| 35. D | 36. C | 37. B | 38. B | 39. D | 40. C | 41. D |
| 42. A | 43. D | 44. | 45. D | 46. D | 47. C | 48. C |
| 49. B | 50. | 51. C | 52. D | 53. A | 54. D | 55. B |
| 56. D | 57. A | 58. | 59. C | 60. A | 61. D | 62. A |
| 63. D | 64. B | 65. D | 66. A | 67. | 68. | |

UNIVERSITY OF PORTHARCOURT 2009/2010 POSTUTME SCREENING EXERCISE TIME: 50MINS

INSTRUCTION: Answer all questions

USE OF ENGLISH

Read this passage carefully and answer the questions that follow.

If present trends continue, the world would face a major crisis by the end of this century. Insufficient, cheap, convenient energy. For without such energy, industrial production will fall, agricultural output will drop, transport will be restricted and *standard of living in developed countries will plummet*. At present, almost all our energy comes from fossil fuels. The earth's reserves of fossil fuels have been formed from organic matter subjected to enormous heat and pressure for millions of years. But such reserves are finite. Because power demand is increasing very rapidly, fossil fuels will be exhausted within a relatively short time. We can estimate the amount of recoverable fuel under the surface of the earth and we know the rate at which it is being extracted. Fairly simple calculations can therefore determine its remaining life. If present trends continue, gas and oil reserves will be exhausted by the middle of the 21st century-about 70 years from now. Similar estimates for coal and wood reserves suggest a projected supply of 250-300 years. Of course long before fossils fuels are exhausted: demand will greatly exceed supply

For too many years, the world has consumed fossil fuels with little thought of the future. Infact, world energy consumption increased almost 600% between 1900 and 1965 and it is projected to increase by another 450% between 1965 and the year 2010, crude oil has been pumped out of the ground for about 10 years, but over half of it has been consumed in the past 18 years. Coal has been mined for over 800 years but over half of it has been extracted in the past 37 years. In sum, most of the world's consumption of energy from fossil fuels throughout history, has taken place within living memory

1. The expression, *standard of living in developed countries will plummet*, means A. the economy of rich nations will stagnate B. economic life will improve in rich nations C. purchasing power will decline sharply in rich nations D. people in developed nations will experience boom
2. The writer warns that the world could A. lose all its oil reserves in a matter of years B. face energy crisis soon if production is not stepped up C. experience scarcity and low energy price soon D. face low energy supply and poor agricultural output

3. *Fossil fuels* as used in the passage include A. wood, kerosene and natural gas B. oil, coal and natural gas C. limited, butane and charcoal D. wood, coal and oil
4. The writer seems to suggest that developed nations should A. always calculate a fossil's fuel's remaining life B. reduce industrial and agricultural production C. reduce dependence on fossil fuels D. review industrial dependence on energy
5. From the writer's description of the world energy situation, we may conclude that A. developing nations will soon experience poverty B. demand for recoverable fuel will plummet C. consumption has not affected production D. decline has not affected demand

In each of the questions 11 – 25, choose the option that best completes the gap(s)

6. Yesterday, my mother asked me A. if I am tired B. are you tired C. whether was I tired D. if I was tired
7. The Bank Manager assured us that there was nothing to worry about ... A. as regard to the loan B. it with regards to the loan C. in regard of the loan D. with regard to the loan
8. The first prize was [A. a carved wooden attractive B. a wooden attractive carved C. an attractive carved wooden D. a carved attractive wooden] tray
9. The class teacher always cautions that difficult jobs should be done A. step from step B. step by step C. steps after steps D. steps by steps
10. I am often angry my brother since he differed. [A. with/to B. about/with C. against/from D. with/with] me often
11. I am writing to you for[A. concerned B. several C. plenty D. myriad of] reasons
12. As we eat the silence, my eyes A. under/looked B. over/surrounded C. in/roamed D. along/observed] the room
13. Is it true that the messenger[A. had returned B. would return C. should return D. should have returned] when the gates were closed
14. The public library has stopped [A. borrowing B. lending C. renting D. loaning] books to readers
15. We won't leave until it [A. will stop B. stopped C. stops D. has stopped] raining
16. The[A. historical B. historicity C. historic D. history] of the event marked the beginning of a new life for the entire cabinet
17. The train[A. was leaving B. has left C. had left D. would leave] before I arrived
18. You can't travel on your own[A. isn't it B. can't you C. won't you D. shan't you]
19. The man is refurbishing the flat with a view to [A. sell B. selling C. have sold D. be selling] it
20. The Secretary-General was shocked at the number of[A. child's soldiers B. childish soldiers C. children soldiers D. child soldiers]
21. In spite of the maid's good looks, her hair is always[A. unkept B. unkempt C. unwashed D. uncared for]
22. By the end of the football match today, we[A. could have known B. might have know C. will have know D. would have known] the best player
23. It is African for a younger person to show[A. understanding B. indifference C. deference D. satisfaction] to elders
24. A nursery rhyme is used to teach pupils how to spell the word[A. hippopotamus B. hippopotamus C. hippopotamus D. hippopotamus]
25. You, who[A. has been B. is C. was D. are] convicted should append

PHYSICS

26. A weight of 1000grams hangs from a lever 20cm to the right of the fulcrum. At the left is a 500gram weight 20cm from the fulcrum, and a 200gram. What is the value of x that will make the lever balanced? A. 50cm B. 20cm C. 10cm D. 30cm E. 70cm

27. What is the relative permittivity of a capacitor if its capacitance with a medium as dielectric is 16farads and its capacitance with vacuum as dielectric is 2 farads.
A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. 2 D. 8 E. 32
28. If the magnification of a virtual image formed of an object 10cm from a convex lens is 3cm. then the focal length of the lens is A. 10cm B. 15cm C. 16cm D. 19cm E. 20cm
29. If a water pump at Kainji Dam is capable of lifting 1000kg of water through a vertical height of 10m in 10s, the power of the pump is. A. 1.0kw B. 10kw C. 12.5kw D. 15.0kw E. 20.0kw ($g = 10\text{m/s}^2$)
30. A simple pendulum with a period of 2.0s has its length doubled. Its new period is
A. 1.00s B. 1.41s C. 0.35s D. 2.83s E. 4.00s
31. In order to find the depth of the sea, a ship sends out a sound wave and receives an echo after one second. If the velocity of sound in water is 1500m/s, what is the depth of the sea?
A. 0.75km B. 1.50km C. 2.20km D. 3.00km E. 3.75km
32. A uniform cylindrical block of wood floats in water with one-third of its height above the water level. In a liquid of relative density 0.8. what reaction of its height will be above the liquid level?
A. $\frac{1}{6}$ B. $\frac{1}{5}$ C. $\frac{1}{3}$ D. $\frac{4}{5}$ E. $\frac{5}{6}$
33. The resistance of a platinum wire at the ice and steam points are 0.75Ω and 1.05Ω respectively. Determine the temperature at which the resistance of the wire is 0.90Ω .
A. 43.0°C B. 50.0° C. 69.9°C D. 87.0°C
34. The electrochemical equivalent of platinum is $5.0 \times 10^{-7}\text{kg/c}$. to plate-out 1.0kg of platinum, a current of 100A must be passed through as appropriate vessel for
A. 5.6hours B. 56hours C. 1.4×10^4 hours D. 2.0×10^4 hours
35. How many grams of water at 17°C must be added to 42kg of ice at 0°C to melt the ice completely. A. 200g B. 300g C. 320g D. 400g
36. The minimum point on the curve $y = x^2 - 6x + 5$ is at A. (1, 5) B. (2, 3) C. (-3, 4) D. (-3, -4) E. (3, -4)
37. In $\sec^2\theta - \tan^2\theta - 3$, then the angle θ is equal A. 30° B. 45° C. 60° D. 90° E. 105°
38. Solve the given equation: $(\log x)^2 - 6\log x + 9 = 0$. A. 27 B. 9 C. $\frac{1}{27}$ D. 18 E. 81
39. Solve the following equations: $4x + 3 = 3x + y$; $3x + y = 2y + 5x - 12$. A. $x = 5, y = 2$ B. $x = 2, y = 5$ C. $x = -2, y = -5$ D. $x = 5, y = -2$ E. $x = -5, y = -2$
40. If $x + 2$ and $x - 1$ are factors of the expressions $lx^3 + 2kx^2 + 24$, find the values of l and k.
A. $l = -6, k = -9$ B. $l = -2, k = 1$ C. $l = -2, k = -1$ D. $l = 0, k = 1$ E. $l = 6, k = 0$
41. Find the equation of the curve which passes through the point (2, 5) and whose gradient at any point is given by $6x - 5$. A. $6x^2 - 5x + 5$ B. $6x^2 + 5x + 5$ C. $3x^2 - 5x - 5$ D. $3x^2 - 5x + 3$
42. A flagstaff stands on the top of a vertical tower. A man standing 60m away from the tower observes that the angles of elevation of the top and bottom of the flagstaff are 64° and 62° respectively. Find the length of the flagstaff. A. $60(\tan 62^\circ - \tan 64^\circ)$ B. $60(\tan 64^\circ - \tan 62^\circ)$ C. $60(\cot 62^\circ - \cot 64^\circ)$ D. $60(\tan 64^\circ - \tan 62^\circ)$
43. Given that $Q = \begin{vmatrix} 6 & 0 \\ 4 & 5 \end{vmatrix}$ and $Q + P = \begin{vmatrix} 7 & -2 \\ 6 & 8 \end{vmatrix}$ Evaluate $[Q + 2P]$
A. 90 B. 96 C. 102 D. 120
44. A function $f(x)$ passes through the origin and its first derivative is $3x + 2$. What is $f(x)$?
A. $y = \frac{3}{2}x^2 + 2x$ B. $y = \frac{3}{2}x + x$ C. $y = 3x^2 + 2x$ D. $y = \frac{1}{2}x$
45. A ship sails a distance of 50km in the direction $S50^\circ E$. Find the bearing of the ship from its original position. A. $S90^\circ E$ B. $N40^\circ E$ C. $S95^\circ E$ D. $N85^\circ E$

CHEMISTRY

46. A certain volume of mass at 295k is heated such that its volume and pressure are now four times the original values. What is the new temperature?
A. 18.6k B. 100k C. 298.0k D. 1192.0k E. 4768.0k

47. The mass of silver deposited when a current of 10A is passed through a solution of silver salt for 4830s is? A. 108.0g B. 54.0g C. 27.0g D. 13.6g [Ag = 108, 1f = 96500C/mol]
48. $2\text{CO}_{(g)} + \text{O}_{2(g)} \rightarrow 2\text{CO}_{2(g)}$. given that $\Delta H [\text{CO}]$ is -110.4kJ/mol and $\Delta H [\text{CO}_2]$ is -393kJ/mol. The energy change of the reaction above is? A. -503.7kJ B. -282.6kJ C. +282.6kJ D. +503.7kJ
49. How many moles of lime stone will be required to produce 5.6g of CaO?
A. 0.20mol B. 0.10mol C. 1.12mol D. 0.56mol
50. One mole of propane is mixed with five moles of oxygen. The mixture is ignited and the propane burns completely. What is the volume of the products at S.T.P?
A. 112.0dm³ B. 67.2dm³ C. 56.0dm³ D. 44.8dm³ [G.M.V = 22.4dm³]

SOLUTION TO UNIPORT 2009/2010 POSTUTME SCREENING EXERCISE

USE OF ENGLISH

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. D | 3. D | 4. A | 5. A | 6. D | 7. D |
| 8. C | 9. B | 10. D | 11. B | 12. A | 13. A | 14. D |
| 15. C | 16. C | 17. B | 18. B | 19. A | 20. C | 21. A |
| 22. D | 23. A | 24. D | 25. A | | | |

PHYSICS

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 26. C | 27. A | 28. B | 29. B | 30. D | 31. A | 32. D |
| 33. B | 34. D | 35. C | 36. E | 37. B | 38. A | 39. A |
| 40. A | 41. A | 42. D | 43. D | 44. A | 45. C | |

CHEMISTRY

- | | | | | |
|-------|-------|-------|-------|-------|
| 46. E | 47. B | 48. A | 49. B | 50. B |
|-------|-------|-------|-------|-------|

UNIVERSITY OF PORTHARCOURT 2008/2009 POSTUTME SCREENING EXERCISE TIME: 50MINS

INSTRUCTION: Answer all questions

Passage 1

Read this Passage Carefully and answer the Questions that follow.

In many places in the world today, the poor are getting poorer while the rich are getting richer, and the programmes of development planning are foreign and appear to be unable to reverse this trend. Nearly all the developing countries have a modern sector, where the patterns of living and working are similar to those in developed countries. But they also have a non-modern sector, where the patterns of living and working are not only unsatisfactory, but in many cases are even getting worse.

What is the typical condition of the poor in developing countries? Their work opportunities are so limited that they cannot *work their way out of their situation*. They are underemployed, or totally unemployed; when they do find occasional work, their productivity is extremely low. Some of them have land, but often too little land. Many have no land, and no prospect of ever getting any. There is no hope for them in the rural areas and so, they drift into the big cities. But there is no work for them in the big cities either and of course no housing. All the same they flock into the cities because their chances of finding some work appear to be greater than in the villages where they are nil. Rural unemployment, then, produces mass migration into the cities; rural unemployment becomes urban unemployment.

The problem can be stated quite simply; what can be done to promote economic growth outside the big cities, in the small towns and villages, which still contain 80 to 90% of the total Population? The primary need is workplaces, literally millions of workplaces,

1. The gap between the rich and the poor widens because there A. are no jobs in the rural areas B. are no employment opportunities in the city C. is no work in the village and the city D. is low growth rate in productivity.
2. The expression *work their way out of their situation* means A. walk from one village to another B. migrate from village to city C. work their way not of Their village D. change their circumstances.
3. Migration to the city among villages is caused by A. attractions of the city B. low productivity in the village C. inadequate job opportunities in the village D. shortage of land for cultivation.
4. Underemployment among the villagers refers to A. lack of sufficient land for everyone B. low productivity when working C. fewer people for many jobs D. more people for fewer jobs.
5. Where are the rich getting richer and the poor poorer? A. in nearly all developing countries B. in a majority of countries in the world C. in developing countries with modern sectors D. in countries with non-modern sectors

Passage II

The Passage below has gaps numbered 6 to 15. Immediately following each gap. four options are provided. Choose the most appropriate option for each gap

The superiority of democracy over other forms of government has long been established. From its Greek origins the 'democracy' has been defined as a political.....6.....[A. pattern B. system C. influence D. hegemony] based on fair representation and liberty for the individual.

The.....7.....[A. principles B. examples C. purposes D. statements] of liberty and equitable representation are rooted in the definition give by Aristotle of Greek about 2500year ago. In the words of Aristotle. "The basis of a democratic.....8.....[A. instinct B. population C. law D. state...]is liberty. Aristotle also emphasized the democratic

.....9[A. issue B. law C. ideal D. form] of 'freedom based on equity' More than 2000 years after Aristotle, the tenants of liberty, justice and equitable representation continue to.,.....10....[A. draw up B. hold sway C. move on D. swing up] from the time of the Greek city states to the emergence of the modern nation-states, the fundamentals of democratic rule have remained universal.....11.....[A. values B. practices C. subjects D, elections]. In the western hemisphere, revolutionary struggles were waged to enthrone democracy over.....12.....[A. liberal B. individual C. autocratic D. collective] rules. Among these were the American Revolution of 1776, the French Revolution of 1789 and the Haitian Revolution of 1804, the first of such projects to be.....13..[A. practiced B. undertaken C. supervised D, introduced] by a black nation.

The American President Abraham Lincoln, in the famous Gettysburg Address.....14.....[A. multiplied B. delineated C. simplified D. complicated] the essential features of democratic.....15.....[A. governance B. government C. notion D. motion] when he defined democracy as a government of the people, by the people, for the people'.

In Questions16to25 choose the word or phrase which best fills the gap in each sentence.

16. The student's unrest resulted..... the expulsion of the ring leaders. A. to B. in C. from D. with E. by.
17. I am very sorry..... to attend the meeting yesterday A. for failure B. in failing C. to having failed D. to fail E. for failing.
18. When you are faced with an examination of this nature Endeavour to keep your mind,..... on the job and not to be distracted for one moment A. at B. in C. for D. on E. to.
19. It all depended on what A. does he want B. he wants C. he does want D. he wanted E. did he want.

20. James reminding that not all that glitters is gold. A. needs B. need C. needing D. needs to E. need to
21. The dull student took a correspondence course as a means.....his standard in the class, A. to improve B. for improving C. of improving D. by improving E. to improving
22. The old politicians were discredited because they tried to.....the people's ignorance A. cash in on B. catch in with C. catch in on D. cash in with E. cash in by
23. That single was enough to spoil lifetime good living A. occurence B. ocurence C. occurrence D. occurrence E. occurrence.
24. Whilst the thief was,.....the passengers, he kept apologizing for the inconvenience he was causing them. A. robbing B. rubbing C. robbing D. rubbing E. robbed.
25. The court ordered the lorry driver to pay for the to my car. A. damages B. heavy damages C. destruction D. many damages E. damage,

PAPER 2: GENERAL SCIENCE ANSWER ALL QUESTIONS

1. A solid weighs 4,8g in air, 2.8g in water and 3.2g in kerosene. The ratio of density of the solid to that of the kerosene is A. 12 B. 3 C. 2 D. $\frac{3}{2}$ E. $\frac{5}{4}$
2. The extension in a spring when 5g wt was hung from it was 0.56cm. If Hooke's law is obeyed, what is the extension caused by a load of 20g wt? a. 1.12cm b. 2.14cm c. 2.52cm d. 2.80cm e. 2.24cm
3. The ice and steam points of a mercury-in-glass thermometer of centigrade scale and of uniform bore correspond respectively to 3cm and 19cm lengths of the mercury thread. When the length is 12cm, the temperature is A. 32°C B. 48°C C. 56°C D. 65° E. 75°
4. An electric cell has an internal resistance of 2 ohms, a current of 0,5A is found to flow when a resistor of 5ohms resistance is connected across it. What is the electromotive force of the cell. A. 5 volts B. 3.5volts C. 2.5 volts D. 1 volt E. 10 volts.
5. A uniform beam HK of length 10m and weighing 200N is supported at both ends. A man weighing 100N stands at a point P on the beam. If the reactions at H and K are respectively 800N and 400N, then the distance HP is A. 4m B. $3\frac{1}{3}\text{m}$ C. 3m D. $6\frac{2}{3}\text{m}$ E. 7m
6. What are the units of thermal conductivity? A. kg.m.sec^2 B. $\text{Joule sec}^{-1}\text{m}^{-1}\text{K}^{-1}$ C. kg.m D. $\text{Newton sec}^{-1}\text{m}^{-1}\text{K}^{-1}$ E. m^{-1}
7. A 500kg car which was initially at rest and traveled with an acceleration of 5ms^{-2} . its kinetic energy after 4 seconds was A. 10^3J B. $2.5 \times 10^3\text{J}$ C. $2 \times 10^3\text{J}$ D. $5 \times 10^3\text{J}$
8. An airforce jet flying with a speed of 335m/s went past an anti aircraft gun. How far is the aircraft 5s later when the gun was fired? A. 8.38m B. 3.350m C. 670m D. 165m E. 67m
9. In a D.C. circuit a 10 microfarad (mf) capacitor is placed in series with a 10 ohm resistor. The total resistance of the combination is A. 10 ohm B. 1 ohm C. zero D. 20ohms E. infinite
10. A potential difference of 6V is used to produce a current of 5A for 200s through a heating coil. The heat produced is A. 4800cal B. 6000cal C. 2400J D. 240kcal E. 6000J
11. Solve the simultaneous linear equation: $2x + 5y = 11$; $7x + 4y = 2$. A. $x = -8, y = 1$ B. $x = -2, y = 4$ C. $x = 2, y = -3$ D. $x = -\frac{34}{27}, y = \frac{73}{27}$ E. $x = \frac{346}{189}, y = -\frac{73}{27}$
12. Solve the system of equation: $2^{x+y} = 32$; $3^{3y-x} = 27$. A. (3, 2) B. (-3, 2) C. (3, -2) D. (-3, -2) E. (2, 2)
13. Solve the equation for all positive values of θ less than 360° , $3\tan\theta + 2 = -1$. A. 135° or 315° B. 45° or 135° C. 315° or 45° D. 315° or 180° E. 360° or 315°
14. Determine the maximum value of $y = 3x^2 - x^3$. A. 0 B. 2 C. 4 D. 6
15. The mean of the numbers 3, 6, 4, x and 7 is 5. Find the standard deviation A. $\sqrt{2}$ B. $\sqrt{3}$ C. 3 D. 2
16. A straight line $y = mx$ meets the curve $y = x^2 - 12x + 40$ in two distinct points. If one of them is (5, 5), find the other. A. (5, 6) B. (8, 8) C. (8, 5) D. (7, 7) E. (7, 5)
17. If $(x - 2)$ and $(x + 1)$ are factors of the expression $x^3 + px^2 + qx + 1$, what is the sum of p and q? A. 0 B. -3 C. 3 D. $-\frac{17}{3}$ E. $-\frac{2}{3}$

18. The ratio of the length of two similar rectangular blocks is 2:3. If the volume of the larger block is 351cm^3 , then the volume of the other block is. A. 234.00cm^3 B. 526.50cm^3 C. 166.00cm^3 D. 729.75cm^3 E. 104.00cm^3
19. At what real value of x do the curves whose equations are $y = x^3 + x$ and $y = x^2 + 1$ intersect. A. -2 B. 2 C. -1 D. 0 E. 1
20. Find the area of a regular hexagon inscribed in a circle of radius 8cm. A. $16\sqrt{3}\text{cm}^2$ B. $96\sqrt{3}\text{cm}^2$ C. $192\sqrt{3}\text{cm}^2$ D. 16cm^2 E. 32cm^2

CHEMISTRY

21. What weight of sodium hydroxide is required to make 500cm^3 of 0.2M solution? (Na = 23, O = 16, H = 1). A. 40g B. 20g C. 10g D. 4g E. 2g
22. 60cm^3 of hydrogen are sparked with 20cm^3 of oxygen at 100°C and 1 atmosphere. The total volume of the residual gasses is A. 60cm^3 B. 10cm^3 C. 40cm^3 D. 30cm^3 E. 70cm^3
23. 0.16g of methane when burnt raises the temperature of 100g of water by 40°C . what is the heat of combustion of methane if the heat capacity of water is $4.2\text{J/g}^\circ\text{C}$. ($\text{CH}_4 = 16$) A. 1.160kJ/mol B. 1.180kJ/mol C. 1.560kJ/mol D. 1.600kJ/mol E. 1.680kJ/mol
24. Hydrated salt of formula $\text{MSO}_4 \cdot \text{XH}_2\text{O}$ contains 45.3% by mass of water of crystallization. Calculate the value of X. A. 3 B. 5 C. 7 D. 10 [M = 56, S = 32, O = 16, H = 1]
25. A given quantity of a gas occupies a volume of 228cm^3 at a pressure of 750mmHg. What will be its volume at atmospheric pressure? A. 220cm^3 B. 225cm^3 C. 230cm^3 D. 235cm^3

SOLUTION TO UNIPORT 2008/2009 POSTUTME SCREENING EXERCISE

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. D | 2. D | 3. C | 4. D | 5. B | 6. B | 7. A |
| 8. D | 9. C | 10. B | 11. B | 12. C | 13. B | 14. C |
| 15. A | 16. B | 17. E | 18. D | 19. D | 20. A | 21. C |
| 22. A | 23. C | 24. A | 25. E | | | |

PAPER 2

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. B | 2. E | 3. C | 4. B | 5. C | 6. B | 7. A |
| 8. D | 9. A | 10. E | 11. D | 12. A | 13. A | 14. B |
| 15. A | 16. B | 17. B | 18. A | 19. E | 20. B | |

CHEMISTRY

- | | | | | |
|-------|-------|-------|-------|-------|
| 21. D | 22. A | 23. E | 24. C | 25. B |
|-------|-------|-------|-------|-------|

UNIVERSITY OF PORTHARCOURT 2006/2007 POSTUTME SCREENING EXERCISE TIME: 30MINS

INSTRUCTION: Answer all questions on the sheet provided

SECTION A

Write a short essay on "the role of an engineer in the society" in the space provided at the reverse of your answer sheet. do not exceed

SECTION B

Read the following passage carefully and answer the questions that follow.

Farming is the most important aspect of agriculture that has attracted attention within the last few years. Agriculture has several other aspects like fishery, livestock and poultry. All these are also important in that they have to do with the production of food items which human being consume for survival. In many parts of the world today, farming has been regarded as the mainstay of the economy. Crops such as cocoa, rubber and cotton have been produced in such commercial quantities that they are sold to other countries. Some countries have better comparative advantage in producing certain farm crops than other countries. In these other countries, there is the need to spend a lot of money on agriculture, particularly farming. Most farmers use outmoded tools. A lot of them have no place to store their crops, most of which are always destroyed by insects and pests before harvest time. All these have adverse effects on their productivity. The government can do a lot to help farmers. Farmers' cooperative societies can be encouraged and loans can be made available to farmers through government institutions. Like banks and Finance Corporation. Farmers can be taught how to build good storage structures for their produce. All these and a lot more can help to improve the conditions of farming in these countries.

1. The most important aspect of agriculture mentioned in the passage is
A. Poultry B. Fishery C. Livestock D. Farming
2. Farming in many countries today is A. An alternative to poultry B. of great assistance to the economy C. For those who are out of jobs D. For the illiterates
3. Some countries produce more and better crops than others because the farmers in the former.
A. are more educated B. have greater manpower C. have more modern equipment D. have more fertile land
4. In order to help improve the state of farming, the government should A. give all farmers enough money to work with B. sell enough fertilizers to all farmers C. find ways of financing and modernizing the farming system D. help farmers with the storage of their crops
5. A lot of crops harvested are wasted because farmers A. allow insects and pests to destroy their crop B. do not have enough money to invest in harvesters C. do not have good storage facilities D. harvest too much at a time

In questions 6 – 15, choose the option opposite in meaning to the underlined word

6. Mr. Jack was most flexible in his instructions. A. Rigid B. Correct C. Stiff D. Upright
7. The University has offered temporary accommodation to its staff. A. Popular B. Permanent C. Recognized D. Regular
8. Mary complained that she slept on the coarse floor A. Smooth B. Rough C. Bad D. Harsh
9. Jim was one of the spectators at the concert A. Ushers B. Judges C. Gests D. Performers
10. The governor declined to give audience to the journalist A. Ignored B. Accepted C. Forgot D. Rejected
11. The debtor's husband is liable for his wife's debts A. Unanswerable B. Responsible C. Unquestionable D. Accountable
12. The lotion recommended by the doctor soothed Okon's aching tooth.
A. Calmed B. Extracted C. Excited D. Worsened
13. The sun cast its shadow on the wall A. Reflection B. Rays C. Resemblance D. Substance
14. He was locked up for a fortnight. A. Released B. Punished C. Remanded D. Locked out
15. The lady acted courageously when thieves attacked her A. Shyly B. Fearlessly C. Indiscreetly D. Timidly

In questions 16-31 choose the option nearest in meaning to the underlined word or phrase.

16. It is claimed that there is an extinct volcano near Jos. A. Extinguished B. Inactive C. Dead D. Disused
17. Bola has a sonorous voice. A. High pitched B. Beautiful C. Strong D. Throaty
18. Some workers went on rampage at a trade fair. A. Turned -violent B. Robber C. Were angry

- D. Demonstrated
19. We have to identify the protagonists of the new movement A. Enemies B. leading figures
C. Opponents D. Believers
 20. My nephew came in stealthily through the back gate A. Briskly B. Boldly C. Wearily D. Quietly
 21. The distance is not more than twenty kilometers as the crow flies A. By the longest route B. By the shortest route C. By air D. By the fastest route
 22. Your extreme patience sometimes infuriates me. A. Impresses B. Annoys C. Frustrates D. Amuses
 23. I cannot understand how he suddenly became audacious contrary to his nature A. Proud B. Bold
C. Rude D. Hostile
 24. His has been a life of make-belief. A. Faith B. Fantasy C. Grandeur D. Religion
 25. Our teacher seldom comes late to school. A. Very often B. Always C. frequently D. Hardly very
 26. At the age of 80, Muse wished he could put the clock back. A. Stop the clock B. Put down the clock C.
Go back in time D. Have more time
 27. The Director left his secret file to the Secretary in good faith. A. With proof B. In anxiety C. In
anticipation D. In trust
 28. The Chief approached the issue with convincing disinterestedness. A. Lack of interest B. Lack of
personal interest C. Inability to be interested D. Unwillingness
 29. It is of fundamental importance that one sleeps properly. A. Undisputed B. Basic C. Special
D. Least
 30. The man insisted on having no strings, attached to the agreement A. Financial obligations
B. Documents C. Conditions D. More obligations
 31. The company has gone under. A. Suffered some loss B. Broken up C. Become broke
D. become bankrupt

For questions 32-36, fill each gap with the appropriate option from the list following the gap

32. Every programming language and software package. . . . [A. Have its B. Have their C. Has its D.
Has their] Limitations.
33. A Programme of good exercise may help a person Fight.[A. Out B. At C. With D. off] cold
34. Three quarters of the Physics class....[A. Improve B. Improves C. are improving D. Is improving]
dramatically
35. A number of students [A. is B. have C. has D. do] Missed the opportunity to reregister
36. A survey of opinions on how pupils feel about teachers... [A. Has been B. Have been C. are being
D. is been] carried out

In questions 37-40 select the option A to D that best explains the information conveyed in the sentence

37. People may pick flowers in this park A. People can pick flowers in this park B. People may not wish
to pick flowers in this park C. People are prohibited from picking flowers in this park D. People
cannot pick flowers from this park
38. Tom ought not to have told me A. Tom did not tell me but he should B. perhaps Tom was wrong to
have told me C. Tom told me but it was wrong of him D. It was necessary for Tom not to tell me
39. Bolade would make a mess of cooking the rice A. It was typical of Bolade to make a
mess of things B. Bolade cannot cook C. Bolade will not cook the rice well D. Bolade-does not like
cooking rice
40. He can't be swimming all day A. it's possible he is not swimming now b. It's very likely he is swimming
now C. He does not have the ability to swim all day D. He would not like to swim all day

PAPER 2

MATHEMATICS

41. The letters of the word MATRICULATION are cut and put in a box. One of the letters is drawn at random
from the box, find the probability of drawing a vowel.
A. $\frac{2}{13}$ B. $\frac{5}{13}$ C. $\frac{6}{13}$ D. $\frac{8}{13}$ E. $\frac{4}{13}$
42. A man invested a total of N50,000 in two companies. If these companies pay dividend of 6% and 8%
respectively. How much did he invest at 8% if the total yield is N3,700.
A. N15,000 B. N29,000 C. N21,400 D. N27,800 E. N35,000

43. Find correct to two decimal places: $100 + \frac{1}{100} + \frac{3}{1000} + \frac{27}{10000}$
 A. 100.02 B. 1000.02 C. 100.22 D. 100.01 E. 100.51
44. In a restaurant, the cost of providing a particular type of food is partly constant and partly inversely proportional to the number of people. If the cost per head for 100 people is 30k and the cost for 40 people is 60k, find the cost for 50 people.
 A. 15k B. 45k C. 20k D. 50k E. 40k
45. The ratio of the length of two similar rectangular blocks is 2:3, if the volume of the larger block is 351cm^3 , then the volume of the other block is. A. 234.00cm^3 B. 526.50cm^3 C. 166cm^3
 D. 729.75cm^3 E. 104.00cm^3
46. Three boys shared some oranges. The first receives $\frac{1}{3}$ of the oranges, the second received $\frac{2}{3}$ of the remainder. If the third boy received the remaining 12 oranges, how many oranges did they share? A. 60
 B. 54 C. 48 D. 42
47. The angle of elevation of the top of a vertical tower 50 meters high from a point X on the ground is 30 degrees. From a point Y on the opposite side of the tower, the angle of the top of the tower is 60degrees. Find the distance between the points X and Y
 A. 14.43m B. 77.73m C. 101.03m D. 115.47m
48. Peter's weekly wages are N20.00 for the first 20 weeks and N36.00 for the next 24 weeks. Find his average weekly wage for the remaining 8 weeks of the year if his average weekly wage for the year is 1430.00 A. 1437.00 B. 1435.00 C. 1430.00 D. N5.00
49. If X is the addition of the prime numbers between 1 and 6 and Y the H.C.F of 6,9,15. Find the production of X and Y/ A. 27 B. 30 C. 33 D. 90
50. A Basket contains green; black and blue balls in the ratio 5: 2:1. If there are 10 blue balls, find the corresponding new ratio when 10 green and 10 black balls are removed from the basket
 A. 1:1:1 B. 4:2:1 C. 5:1: D. 4:1:1

PHYSICS

51. Which of the following statement about liquid pressure is NOT correct? The pressure
 A. At a point in a liquid is proportional to the depth B. At any point in a liquid is the same at the same level C. is exerted equally in all directions at any point D. of a liquid at any point of its container acts in a direction perpendicular to the wall E. At a particular depth depends on the shape of the vessel.
52. A ship traveling towards a cliff receives the echo of its whistle after 3.5 seconds. A short while later, it receives the echo after 2.5 seconds. If the speed of sound in air under prevailing conditions is 250m/s how much closer is the ship to the cliff? A. 10m B. 125m C. 175m D. 350m E. 1000m
53. A force of 16N applied to a 4.0kg block that is at rest on a smooth horizontal surface. What is the velocity of the block at $t = 5\text{seconds}$? A. 4m/s B. 10m/s C. 20m/s D. 50m/s E. 50m/s
54. The distance traveled by a particle starting from rest is plotted against the square of the time elapsed from the commencement of motion. The resulting graph is linear. The slope of this graph is a measure of A. initial displacement B Initial velocity c. Acceleration D; Half the acceleration E. Half the initial velocity
55. A small needle is carefully floated on water in a beaker. When a few drops of kerosene are introduced into the water, the needle sinks. Which of the following statements correctly explain(s) the observation?
 i. There is tension on the water surface
 ii. Kerosene reduces the density of water so that the needle becomes denser than water
 iii. Kerosene reduces the surface tension of water
 A. I only B. II only C. III only D. I and II E. I and III
56. Which of the following is a correct explanation of the inertia of a body? A. Ability to overcome the earth's gravity B. Reluctance to stop moving C. Readiness to start moving D. Reluctance to start moving and its readiness to stop moving once it has begun to move E. Reluctance to start moving and its reluctance to stop moving once it has begun to move.
57. Two masses 40g and 60g are attached firmly to the ends of a light meter rule. The centre of gravity of the system is A. At the midpoint of the meter rule B. 40cm from the lighter mass

- C. 40cm from the heavier mass D. 60cm from the heavier mass
58. The mode of heat transfer which does not require a material medium is A. Conduction
B. Radiation C. Convection D. Propagation
59. When two objects P and Q are supplied with the same quantity of heat, the temperature change in P is observed to be twice that of Q. the mass of P is half of Q, the ratio of the specific heat capacity of P to Q is A. 1:4 B. 4:1 C. 1:1 D. 2:1
60. Which of the following conditions will make water boil at a temperature of 100°C when the atmospheric pressure is 750mm Hg. A. Increase the external pressure B. Reduce the external pressure C. Heat more rapidly at the same pressure D. Reduce the external pressure by a quarter

CHEMISTRY

61. Stainless steel is an alloy of: A. Carbon, Iron and lead B. Carbon, Iron and chromium
C. Carbon, Iron and copper D. Carbon, Iron and silver E. Carbon and Iron only
62. Crude petroleum usually seen on some Nigeria's creeks and water ways can be dispersed or remove by
A. heating the affected parts in order to boil off the petroleum B. Mechanically stirring to dissolve the petroleum in water C. Pouring organic solvents to dissolve the petroleum
D. Spraying the water with detergents E. Cooling to freeze cut the petroleum.
63. Solutions X, Y and Z have P^H values of 3.0, 5.0 and 9.0 respectively. Which of the following statements is correct? A. All the solutions are acidic B. All solutions are 'basic C. Y and Z are more acidic than water
D. Y is more acidic than X E. Z is the least acidic
64. A molar solution of caustic soda is prepared by dissolving A. 40g NaOH in 100g of water
B. 40g NaOH in 1000g of water C. 20g NaOH in 500g of solution D. 20g NaOH in 1000g of solution E. 20g NaOH in 80g of solution
65. Water is said to be hard if it A. Easily forms ice B. Has to be warmed before sodium chloride dissolves in it
C. Forms an insoluble scum with soap D. Contains nitrates E. Contains sodium ions.
66. Chemical substances are mentioned as follows (i) sour taste (ii) slippery to touch (iii) yields alkaline gas with ammonium salts (iv) has p^H less than 7 (v) turns phenolphthalein pink. Which of the above are NOT typical properties of alkali? A. (i), (iv) and (v) B. (iv) and (v) C. (i) and (iv) D. (ii) and (v) E. (ii), (iii) and (v)
67. The movement of liquid molecules from the surface of the liquid into gaseous phase above it is known as A. Brownian movement B. Condensation C. Evaporation D. Liquefaction
68. Ammonia gas is normally dried with A. Concentrated sulphuric acid B. Quicklime
C. Anhydrous calcium chloride D. Magnesium sulphate
69. Oxygen demanding wastes are considered to be a water pollutant because they:
A. Deplete oxygen which is necessary for the survival of aquatic organisms B. Increase oxygen which is necessary for the survival of aquatic organisms C. Deplete other gaseous species which are necessary for the survival of aquatic organisms
70. Mortar is NOT used for underwater construction because
A. It hardens by loss of water
B. its hardening does not depend upon evaporation
C. It requires concrete to harden
D. It will be washed away by the flow of water

SOLUTION TO UNIPORT 2006/2007 POSTUTME SCREENING EXERCISE

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. D | 2. B | 3. C | 4. C | 5. C | 6. A | 7. B |
| 8. A | 9. D | 10. B | 11. A | 12. D | 13. D | 14. A |
| 15. D | 16. C | 17. A | 18. A | 19. B | 20. D | 21. B |
| 22. B | 23. B | 24. B | 25. D | 26. D | 27. D | 28. A |
| 29. B | 30. C | 31. D | 32. C | 33. D | 34. B | 35. C |
| 36. A | 37. A | 38. C | 39. B | 40. A | | |

MATHEMATICS

- | | | | | | | |
|-------|-------|-------|-------|-----|-----|-----|
| 41. C | 42. B | 43. A | 44. A | 45. | 46. | 47. |
| 48. | 49. | 50. | | | | |

PHYSICS

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 51. E | 52. B | 53. C | 54. D | 55. E | 56. E | 57. C |
|-------|-------|-------|-------|-------|-------|-------|

58. B

59. B

60. A

CHEMISTRY

61. B

62. D

63. E

64. B

65. C

66. C

67. C

68. B

69. A

70. C